Kouvola–Kotka Corridor Finland

Gate 1 Infrastructure & Investment Due Diligence Dossier

CANDIDATE ID
EU-FI-02

RESEARCH DATE
August 2026

GATE 1 RECOMMENDATION
FAIL

Geography assessed	Kouvola, Kotka, Hamina and the wider Kymenlaakso region, South-East Finland
Question	Could this corridor realistically support a new 100–500+ MW AI infrastructure campus, and is there enough evidence to justify active site origination and operator engagement by Atlas?
Answer	**The corridor can support such a campus. Atlas cannot originate it.** The region is a validated, capital-saturated hyperscale market in which every material site, grid position and heat off-take relationship has already been taken by better-capitalised incumbents.
Research status	Desk research only. No site visit, no counterparty contact, no Fingrid enquiry, no land control, no operator engagement.
Evidence base	93 catalogued sources; 116 numbered evidence records (key records reproduced in Appendix A); 4 material contradictions disclosed; 6 gate-blocking unknowns identified.

Status of this document. This is a Gate 1 screening dossier produced from public-domain evidence between research cut-off and August 2026. It contains no confidential information and no information obtained from any counterparty. Every quantitative claim carries a source reference in Appendix B. Conclusions are classified by evidence strength throughout; readers should not treat an INDICATED or UNVERIFIED finding as a basis for capital commitment.

Document Control & Reading Guide

Field	Value
Document	Project_Atlas_Kouvola_Kotka_Gate1_Full_Dossier.pdf
Programme	Project Atlas — European AI Infrastructure Site Origination, Albedo Industries
Candidate	EU-FI-02 — Kouvola–Kotka corridor, Kymenlaakso, Finland
Gate	Gate 1 — Infrastructure & Investment Screening
Methodology	Falsification-first. The dossier is structured to disprove the investment thesis; it advances only if disproof fails.
Prepared	August 2026
Classification	Internal — Investment Committee
Supersedes	None. First issue for this candidate.
Companion candidate	EU-ES-01 (As Pontes de García Rodríguez, Galicia) — Gate 1 issued separately

Evidence classification used throughout

Class	Meaning	Use in this dossier
CONFIRMED	Directly supported by authoritative primary evidence.	Government legislation, Fingrid publications, company investor announcements, official statistics.
STRONGLY INDICATED	Multiple credible sources agree; one critical confirmation outstanding.	Campus capacities reported consistently by trade press and the developer but not independently verified.
INDICATED	Credible evidence exists but is incomplete.	Regional fibre presence inferred from a carrier's stated network-building principle.
UNVERIFIED	Plausible hypothesis requiring validation.	Water availability at specific plots; site-level soil conditions.
UNKNOWN	No adequate evidence identified.	Node-level available MW; connection queue position; DSO headroom.
BLOCKER	Evidence identifies a potentially material impediment.	Transmission connection restriction; electricity tax reclassification; land pre-emption.

Prohibited inferences

Atlas methodology forbids the following silent transformations. Each is flagged in this document wherever a source attempts it: **infrastructure proximity → infrastructure availability**; **transmission voltage → available MW**; **industrial land → controllable land**; **fibre presence → hyperscale redundant connectivity**; **renewable generation → obtainable PPA**; **operator activity in Finland → operator demand for this site**; **municipal enthusiasm → permitting certainty**. Regional marketing material for this corridor commits at least four of these seven. See §13 and §39.

Contents

PART I

Investment Case

2. Executive Summary

Kouvola–Kotka is a real hyperscale location. That is precisely why it is not an Atlas opportunity. Between December 2023 and July 2026 the corridor absorbed a wave of capital — Google, atNorth, Hyperco/Edgnex, DayOne, ByteDance, Novagen — that has already claimed the brownfield estate, the substation adjacencies and the district-heating off-take relationships that Project Atlas would have needed to originate a site. Atlas arrives four years late to a market it would have had to enter in 2022.

Why Kouvola–Kotka? Why Finland? Why now?

Why Finland. Finland offers a near-decarbonised, low-priced, high-reliability electricity system in a NATO/EU jurisdiction with a cold climate. Nuclear supplies roughly 40% of generation following the commissioning of the 1,600 MW Olkiluoto-3 reactor, hydro roughly a fifth, wind a fast-growing further fifth. [FI02-PWR-004] Finland is a single bidding zone, which removes intra-national locational price risk entirely. The Finnish day-ahead average fell from roughly €46/MWh in 2024 to roughly €41/MWh in 2025, among the cheapest in Europe, with 447 negative-price hours in 2025. [FI02-PWR-011, -012] Fingrid forecasts national consumption rising from 86 TWh in 2025 to 103–123 TWh by 2030 and as high as 159 TWh by 2035, driven principally by data centres. [FI02-PWR-002]

Why the corridor. Kymenlaakso is a classic just-transition geography: a pulp-and-paper region whose industrial base contracted sharply, leaving behind heavy electrical infrastructure, large levelled brownfield plots, cooling water, district-heating networks and a workforce with process-industry skills. Stora Enso closed the Sunila pulp mill in Kotka and one of two paper machines at Anjala; the Finnish Government granted Kymenlaakso dedicated structural-change support in 2023. [FI02-MKT-014] This is the same conversion logic that produced Google's Hamina campus, built on the Summa paper mill bought from Stora Enso in 2009. [FI02-MKT-001]

Why now — and why that is the problem. The conversion thesis is correct, validated, and executed by others. The table below is the single most important object in this dossier.

Development	Location	Scale	Status as at August 2026
Google Hamina campus	Hamina (Summa)	Undisclosed; 7th facility	Operating since 2011. Cumulative investment stated above €4.5bn including a €1bn expansion. 50 ha adjacent land bank acquired 2022. Heat recovery to Haminan Energia covering c.80% of Hamina's district-heat demand. CONFIRMED
atNorth FIN04	Myllykoski, Kouvola	430 MW target, 45 ha	Under construction. YIT design-build contract worth c.€300m agreed July 2026, completion end-2027. Permit application filed 2026 for three further buildings at Ummeljoki/Myllykoski. CONFIRMED
Hyperco / DayOne (ByteDance–TikTok)	Koria, Kouvola	€1bn; 24 ha; c.28,000 m²	Under construction (YIT main contractor; Caverion MEP; Eltel 110 kV substation and cabling to the Fingrid substation, €16m). Completion May 2027. TikTok confirmed as primary tenant under Project Clover. CONFIRMED
Hyperco Koria second building	Koria, Kouvola	100 MW; 56,670 m²	Building permit application before the Kouvola Technical Committee June 2026; early-works request submitted. STRONGLY INDICATED
Novagen Data Centre	Voikkaa, Kouvola	c.21 ha; €48m	Preliminary agreement signed 20 February 2026 to buy the former UPM Voikkaa paper mill estate from Redeve; €200,000 deposit paid. Chairman also chairs Hyperco. STRONGLY INDICATED
Further Kouvola sites	Kuusankoski; Ummeljoki	Not disclosed	Yle reported in March 2026 that data centres are under construction or in preparation at Myllykoski, Ummeljoki, Koria and Kuusankoski. STRONGLY INDICATED

Table 1. The corridor's committed and in-permitting data centre pipeline. Six developments across one city of 77,625 people and one adjacent coastal town. Sources: §21, Appendix E.

What infrastructure already exists

- **Transmission.** Fingrid's Koria 400/110/20 kV substation sits inside the Kouvola municipal area and is the corridor's transmission anchor, rebuilt 2016–2019. The Kymi 110 kV substation north of Kotka was expanded in 2020 to accept a customer's new substation. [FI02-GRD-001, -003]
- **Heat networks.** Three municipal utilities operate district-heating systems with declared data-centre heat-reuse intent: KSS Energia (Kouvola), Kotkan Energia (Kotka), Haminan Energia (Hamina). [FI02-HEA-001..003]
- **Port and logistics.** HaminaKotka is Finland's largest universal and container port, roughly 1,100–1,200 ha, direct E18 motorway and rail access, and explicit project-cargo capability. [FI02-LOG-001] Kouvola is Finland's principal railway junction. [FI02-LOG-004]
- **Cooling.** The Gulf of Finland at Kotka/Hamina supports seawater cooling — proven at scale by Google since 2011. [FI02-WTR-001] Inland, the Kymijoki river system serves Kouvola.

What remains unverified — and what kills it

PRIMARY BLOCKER

Fingrid restricted the connection of new large electricity consumption to the main grid in southern Finland for the period 2025–2027, pending completion of reinforcement investments. A Fingrid unit manager stated publicly that no new customers on the west coast or in southern Finland could be connected to the main grid until 2027 at the earliest. Reported thresholds for the associated approval procedure begin at 2–10 MW in southern Finland, varying by region. Fingrid has separately disclosed enquiries totalling around 500,000 MW of production and consumption connection interest against a national peak consumption of roughly 15,000 MW. [FI02-GRD-006, -007, -008] BLOCKER

Kymenlaakso lies within southern Finland. Atlas has **not** established whether the restriction binds at the specific nodes serving Kouvola and Kotka, nor what capacity, if any, remains uncommitted after the six developments in Table 1. That is the central unresolved fact of this dossier.

AVAILABLE MW NOT VERIFIED

No public source quantifies uncommitted demand-connection capacity at Koria, Kymi, or any other node in the corridor. Atlas has found no capacity map, queue disclosure, or reinforcement plan that answers the question at node level. **Any statement in marketing material that a corridor site offers ">100 MW long term" is an unverified vendor claim, not a capacity allocation.** UNKNOWN

NO OPERATOR DEMAND VALIDATED

No AI infrastructure operator has expressed interest in an Atlas-originated site in this corridor. Atlas is at **Level 0 — no contact** on the demand-validation ladder in §34. The operator activity documented in Table 1 is demand for *other developers' sites* and must not be represented as demand for Atlas.

The second blocker: the tax base moved against the thesis in July 2026

Finland removed the data-centre electricity tax concession with effect from **1 July 2026**. Electricity used in data centres moved from the lower tax category II to the general category I — from 0.05 c/kWh to 2.24 c/kWh, an increase of 2.19 c/kWh. The Government estimated the reform would raise electricity tax revenue by €47m annually at 2026 levels. [FI02-REG-001] CONFIRMED

A replacement state aid scheme — a fixed-term ten-year tax relief based on electricity consumed, capped at a maximum €30m annual revenue loss and conditional on registration plus waste-heat reuse or above-baseline energy efficiency — was decided by the Ministerial Committee on Economic Policy in December 2025, with legislation entering into force on the same date. [FI02-REG-003] The detailed terms, eligibility and EU state aid compatibility were still being specified at the time of that announcement.

The gross arithmetic matters. At 2.19 c/kWh, a 100 MW facility at high load factor incurs on the order of **€19m per year** in additional electricity tax before any relief. Over a ten-year horizon that is a nine-figure change in the operating cost base of a single campus, arriving in the middle of the corridor's construction wave. Google publicly indicated that future Finnish expansion decisions depend on the tax settlement, and XTX Markets was reported to be reviewing a Kajaani investment exceeding $1.1bn in response. [FI02-REG-006]

What scale might theoretically be possible

If — and only if — connection capacity were shown to exist, the corridor's physical envelope would support 100–500 MW. atNorth's own 430 MW target across 45 ha at Myllykoski establishes that the land, heat off-take and grid architecture can carry that order of magnitude in this specific geography. [FI02-MKT-004] The constraint is not physics or planning. It is that this envelope is being consumed by others, in real time, with signed EPC contracts.

Investment thesis, counter-thesis, and the next action

Investment thesis	A just-transition industrial corridor with 400 kV transmission, brownfield land, seawater and river cooling, district-heat off-take and Finland's largest port offers a repeatable, low-cost route to a 100–500 MW European AI campus, with Google's twenty-year presence de-risking the location for operators.
Counter-thesis	Every element of that thesis is true and has therefore already been acted upon. Six developments are committed or in permitting across the corridor. The transmission connection window is administratively closed until at least 2027. The fiscal advantage that drew the first wave was withdrawn on 1 July 2026. Atlas holds no land, no queue position, no capital and no operator relationship, and has no identified source of differentiation against Partners Group, Damac/Edgnex, Brookfield-backed DayOne or Alphabet.
Primary blocker	Fingrid's 2025–2027 restriction on new large demand connections in southern Finland, compounded by unquantified node-level headroom after six committed projects. [FI02-GRD-006]
Primary opportunity	Not site origination. The only surviving hypothesis is a *non-origination* role — see §46. Atlas has not tested it and it is out of scope at Gate 1.
Next critical action	None in this corridor. Reallocate the 30-day validation budget to a candidate where Atlas can plausibly hold a differentiated position. The one question that would reverse this decision is stated in §46.

3. Gate 1 Decision

GATE 1 RECOMMENDATION · CANDIDATE EU-FI-02

FAIL

Do not progress to Gate 2. Do not commence outreach. Do not allocate the 30-day validation programme.

Explanation of the decision

Atlas separates two questions that regional marketing material routinely conflates: *is this a good place to build a data centre?* and *can Atlas build one here?* The answer to the first is yes. The answer to the second is no, and the second is the only one that governs capital allocation.

The FAIL rests on four independent findings, any two of which would be sufficient.

#	Finding	Class	Consequence for Atlas
1	Land and brownfield estate substantially pre-empted	CONFIRMED	The two largest convertible estates — Myllykoski (Redeve → atNorth, 21 ha sold, >50 ha retained) and Voikkaa (Redeve → Novagen, c.21 ha under preliminary agreement at €48m) — are controlled or contracted. Koria is built out by Hyperco. The Business Finland site catalogue entries for Kotka Mussalo and Kouvola Savontalo carry reservation flags or stale dating. Atlas would be bidding against Damac and Partners Group for residual parcels. See §13–14.
2	Transmission connection administratively restricted	BLOCKER	New large demand connections in southern Finland are restricted through 2027 pending reinforcement. Node-level headroom in Kymenlaakso after six committed projects is unquantified and unquantifiable from public sources. A greenfield Atlas project would enter a queue behind everything in Table 1. See §7.
3	Fiscal basis changed against the thesis on 1 July 2026	CONFIRMED	The 2.19 c/kWh electricity tax increase is in force. The offsetting state aid scheme is capped at €30m nationally, is conditional, and its detailed terms and EU state aid clearance were unresolved at the point of decision. Incumbents with signed EPC contracts absorb this; a marginal new entrant underwriting today does not. See §27.
4	No demand, no capital, no differentiation	CONFIRMED	Atlas is at demand-validation Level 0. Competing developers are backed by Partners Group, Damac, Brookfield and a sovereign wealth investor, one of whom closed a €1bn holdco financing for DayOne. Atlas has identified no asset, relationship or insight that these parties lack. See §23, §34.

What would move the candidate upward

Only one thing, and it is not incremental research. Atlas would need **written confirmation from Fingrid of uncommitted demand-connection capacity of ≥100 MW at a named node in Kymenlaakso, together with an identified parcel of ≥20 ha under Atlas's exclusive option.** Both conditions, simultaneously, in writing. Absent either, further work in this corridor is a sunk cost. Atlas assesses the probability of both conditions being satisfiable as low, because the second is a market transaction Atlas cannot win against the incumbents identified in §23.

What would move the candidate downward — to permanent rejection

- Confirmation from Fingrid that Kymenlaakso node capacity is fully committed through the reinforcement horizon.
- Completion of the Novagen–Redeve Voikkaa transaction, removing the last large uncontracted brownfield estate.
- Publication of state aid terms that exclude speculative or merchant-model campuses from relief eligibility.

What would have been required to reach Gate 2

Gate 2 in the Atlas framework requires site control, a written grid position and a Level 3 demand signal. This candidate satisfies none of the three and has no credible pathway to any of them within twelve months. The gap is not evidentiary — it is structural.

A note on what a FAIL means

A FAIL at Gate 1 is a successful outcome of the programme, not a failure of it. Atlas exists to eliminate weak candidates as efficiently as it identifies strong ones. This dossier cost desk research and no capital. The counterfactual — twelve months of outreach into a market where the land was gone before the first email — would have cost considerably more. The corridor should be retained in the Atlas database as a *reference geography*: it is the clearest available worked example of how a just-transition industrial region converts to AI infrastructure, and its sequence of events is directly instructive for candidates at an earlier stage of the same cycle.

4. Candidate Scorecard

Atlas scores every candidate twice. **Opportunity Score** asks how attractive the candidate would be if all assumptions proved true. **Evidence Confidence** asks how much Atlas actually knows today. A high opportunity score with low evidence confidence is a research target. A high opportunity score with high evidence confidence held by *someone else* — which is this candidate — is not an opportunity at all. The distinction is the whole point of the exercise.

#	Dimension	Opportunity (0–5)	Evidence Confidence (0–5)	Reasoning
1	**Power**	4	1	Clean, cheap, single-bidding-zone electricity with 400 kV presence at Koria. But available MW is unknown and southern Finland is connection-restricted to 2027. High potential, near-zero knowledge.
2	**Fibre**	3	1	Cinia builds along railways and Kouvola is Finland's main junction, which makes presence likely. But C-Lion1 lands at Helsinki and Hanko, *not* in this corridor. Redundant hyperscale-grade diverse routing is unproven.
3	**Land**	4	4	Excellent brownfield estate — and Atlas knows with high confidence that it is taken. This is the rare case where high evidence confidence produces a negative score contribution.
4	**Water / Cooling**	4	2	Seawater cooling proven at Hamina since 2011; Kymijoki inland. Plot-level abstraction rights, discharge permits and municipal capacity all unverified.
5	**Climate**	5	4	Among the best free-cooling climates in Europe. Well documented. No material downside for high-density AI racks. The one genuinely unambiguous strength.
6	**Construction**	4	4	YIT, Caverion, Eltel all demonstrably delivering data centre work in this exact corridor. Capability confirmed — but that capacity is now contracted to competitors through 2027.
7	**Regulation**	3	3	Stable rule-of-law jurisdiction, but the sector-specific regime moved adversely on 1 July 2026 and the replacement aid scheme is unresolved in detail. Zoning amendment risk is material where a plot is not already zoned for the use.
8	**Customer demand**	4	0	Real operator demand exists in the corridor — for other people's sites. Atlas has zero evidence of demand for an Atlas site. Level 0.
9	**Community**	4	3	Municipal leadership publicly welcoming; structural-change region actively seeking investment. Note the prohibited inference: enthusiasm is not permitting certainty.
10	**Competition**	1	5	The worst score in the matrix and the best-evidenced. Six developments, four sponsors, tier-one capital, signed EPC contracts. Atlas has no answer to this.
11	**Capital environment**	4	3	Abundant institutional capital is flowing into Finnish data centres. It is flowing to sponsors with track record; Atlas has none in this asset class.
	Weighted mean	**3.6**	**2.7**	**An attractive geography that Atlas cannot access.**

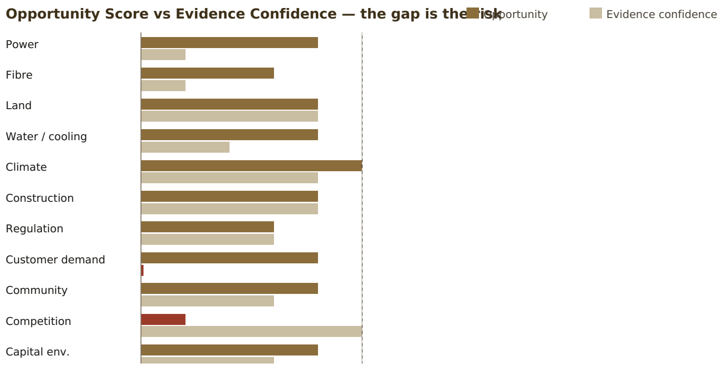


Figure 1. Two dimensions where wide gaps signal research targets — except Competition and Customer Demand, where the evidence is complete and the news is bad. Red bars mark scores that argue against the candidate.

How to read this scorecard

Rows 1, 2, 4 and 8 have wide opportunity-to-evidence gaps: on a normal candidate these would define the research programme. Rows 3 and 10 are the decisive ones. On **Land**, Atlas has high confidence and the answer is adverse. On **Competition**, Atlas has maximum confidence and the answer is adverse. When the two best-evidenced dimensions both argue against, additional research on the poorly-evidenced dimensions cannot change the recommendation — it can only refine the reasons. That is why this candidate fails at Gate 1 rather than moving to "continue researching".

Power & Energy

Power is the only variable that can independently kill this candidate, and it is the variable Atlas knows least about. This section establishes what is confirmed about the Finnish system, what is confirmed about the corridor's transmission assets, and — most importantly — the precise boundary at which confirmed knowledge stops and vendor assertion begins.

5. The Finnish Electricity System

Generation mix and self-sufficiency

Finland's power system changed structurally in 2023 with the commissioning of Olkiluoto-3, a 1,600 MW reactor and Europe's largest. Together with the older Olkiluoto units and Loviisa — the latter located approximately 60 km west of Kotka — nuclear now covers on the order of 40% of Finnish generation. Hydro from the northern Kemi and Oulu river systems contributes roughly a fifth to a quarter, wind a comparable and fast-growing share, bioenergy from the forest industry roughly a tenth, and solar a small but rapidly expanding fraction. [FI02-PWR-004, -005] STRONGLY INDICATED

The practical consequence for a data centre developer is a carbon intensity that is exceptionally low by European standards. Fingrid reported the emission factor for electricity consumed in Finland at 32 gCO2/kWh for the first half of 2026, against 30 gCO2/kWh in the comparable 2025 period. [FI02-PWR-006] CONFIRMED For an operator with a 24/7 carbon-free energy target, Finland is close to a solved problem on an annual-matching basis.

Market structure and the single bidding zone

Finland is a single bidding zone. This is a material and under-appreciated feature. In markets segmented into multiple zones — Sweden, Norway, Italy — siting decisions carry embedded locational price risk that can dominate the economics of a twenty-year asset. In Finland, a campus at Kouvola faces the same day-ahead price as a campus in Helsinki or Oulu. Fingrid has explicitly cited maintaining Finland as a single price area as an objective of its transmission investment programme. [FI02-PWR-013]

The corollary is that **location within Finland is a pure infrastructure and cost-of-delivery decision, not an energy-price arbitrage.** Any pitch for this corridor that leans on "cheap Finnish electricity" is making a national argument, not a corridor argument, and does not differentiate Kouvola–Kotka from Lahti, Mäntsälä, Lappeenranta, Kajaani or Rovaniemi — all of which host competing projects.

Prices, volatility and negative hours

Metric	Value	Note
Annual average day-ahead, 2014–2020	€28–47/MWh	Historic baseline before the European energy shock. [FI02-PWR-010]
Annual average day-ahead, 2021–2024	€46–154/MWh	Range across the four years; 2022 the outlier. [FI02-PWR-010]
Annual average, 2024	c.€46/MWh	[FI02-PWR-011]
Annual average, 2025	c.€41/MWh	Fell year-on-year while much of Europe rose. [FI02-PWR-011] STRONGLY INDICATED
Negative-price hours, 2024	724	[FI02-PWR-012]
Negative-price hours, 2025	447	Down 38% year-on-year. [FI02-PWR-012]
Record hourly high	€1,896/MWh	Early 2024. [FI02-PWR-010]
Record hourly low	−€500/MWh	24 November 2023. [FI02-PWR-010]
Business retail, all-in, Dec 2025	c.€0.091/kWh	Includes power, distribution, transmission and all taxes and fees — i.e. a delivered price, not a wholesale price. [FI02-PWR-014]

Table 2. Finnish price environment. Note the spread: a 40-fold range between record low and record high hours. Volatility, not level, is the design constraint.

The pattern that matters for AI infrastructure is **increasing wind penetration producing both deeper troughs and sharper spikes.** A campus with genuine load flexibility can harvest the troughs; a campus running fixed 24/7 training workloads at 95% utilisation captures the annual average and eats the spikes. The 2025 fall in negative hours also signals that the arbitrage window is narrowing as storage and flexible load enter the market — Fingrid has received more than 30,000 MW of grid energy storage connection requests, roughly double Finland's all-time peak consumption. [FI02-PWR-015]

Fingrid, DSOs, and the transmission/distribution boundary

Fingrid Oyj operates the Finnish transmission system (400 kV, 220 kV and much of the 110 kV backbone) and holds statutory system responsibility. Distribution below that level is operated by regional DSOs; in Kymenlaakso the relevant regional players include Kymenlaakson Sähköverkko and the municipal utilities. The connection route for a 100 MW+ campus is **direct to Fingrid** in most realistic architectures, which matters because the technical requirements and restrictions discussed below apply specifically to Fingrid's direct connectees.

Fingrid is investing at a historically unprecedented rate. The 2026–2035 main grid development plan provides for approximately **€5.2 billion** of investment, up from around €4bn in the prior plan, with €1.7bn falling in 2025–2028 alone. [FI02-GRD-010, -011] CONFIRMED Fingrid has contracted Hitachi Energy for seven 400 MVA / 400 kV power transformers with an option for two more, manufactured in Vaasa. [FI02-GRD-012]

Demand trajectory

Fingrid's September 2025 estimate projects Finnish consumption rising from **86 TWh in 2025 to 103–123 TWh by 2030**, reaching **159 TWh by 2035** in the strongest scenario. The stated principal drivers are new industry and, above all, new data centres. [FI02-PWR-002] CONFIRMED A separate Fingrid forecast places industrial and data centre consumption alone rising from 40 TWh in 2025 to 101 TWh in 2035. [FI02-PWR-003]

Actual 2026 outturn is running ahead of trend: consumption in January–June 2026 was 46.1 TWh, up 6.3% year-on-year, partly weather-driven. [FI02-PWR-007] Demand is arriving faster than the network is being reinforced. This is the mechanism that produced the connection restriction examined in §7.

6. Kouvola-Kotka Grid Topology

Atlas has identified the following corridor transmission assets from Fingrid primary sources. This register is **incomplete**: Fingrid does not publish a consolidated public asset schedule with capacities, and Atlas has deliberately not estimated ratings that are not published.

Asset	Voltage	Location	Operator	Function	Evidence
Koria substation	400 / 110 / 20 kV	Koria, Kouvola	Fingrid	The corridor's transmission anchor. Serves reliability of supply for Kymenlaakso and Päijät-Häme.	Fingrid project page. Originally built in the 1970s; a near three-year refurbishment ran from November 2016, completing end-2019, replacing the 400 kV switchgear with a full-duplex arrangement using disconnecting circuit-breakers, and largely renewing 110 kV primary equipment. [FI02-GRD-001] CONFIRMED
"Rautarouva" line	Not stated	Koria ↔ Yllikkälä (Lappeenranta)	Fingrid	East–west corridor link toward South Karelia.	Renewed under a separate project alongside the Koria refurbishment. [FI02-GRD-002] CONFIRMED
Kymi substation	110 kV (switchgear expanded)	North of Kotka	Fingrid	Connects industrial load north of Kotka to the main grid.	Expanded in 2020 with two new transmission line bays specifically to connect a customer's new substation; the customer built the connecting lines. [FI02-GRD-003] CONFIRMED
Kymi line arrangements	Not stated	Kotka	Fingrid	Line reconfiguration project ("Kymin johtojärjestelyt").	Listed in Fingrid's project archive. Scope and dates not established by Atlas. [FI02-GRD-004] INDICATED
Kyminlinna–Pernoonkoski	Not stated	Kotka area	Fingrid	Transmission line project.	Listed in Fingrid's project archive. Scope and dates not established. [FI02-GRD-005] INDICATED
Hyperco Koria customer substation	110 kV	Koria, Kouvola	Customer-owned	Dedicated data centre connection.	Eltel turnkey contract, c.€16m, for a 110 kV substation plus cabling from the Hyperco site to the Fingrid substation. Construction from summer 2025, Eltel scope planned complete 2026. [FI02-GRD-009] CONFIRMED
Hikiä–Toivila	Not stated	Southern Finland (outside corridor)	Fingrid	Reinforcement that increases connection capacity in southern Finland.	Completion cited as 2029; explicitly identified as the trigger for lifting the extended storage-connection restriction. [FI02-GRD-013] CONFIRMED
Eastbound 400 kV link	400 kV	Eastern Finland	Fingrid	Planned; EIA being launched.	Fingrid stated in September 2025 that it is launching an environmental impact assessment in preparation for implementing an eastbound 400 kV transmission link, and is monitoring Eastern Finland as customer projects progress. [FI02-GRD-014] CONFIRMED

Table 3. Corridor grid infrastructure register (partial). See Appendix C for the full record including uncertainty notes.

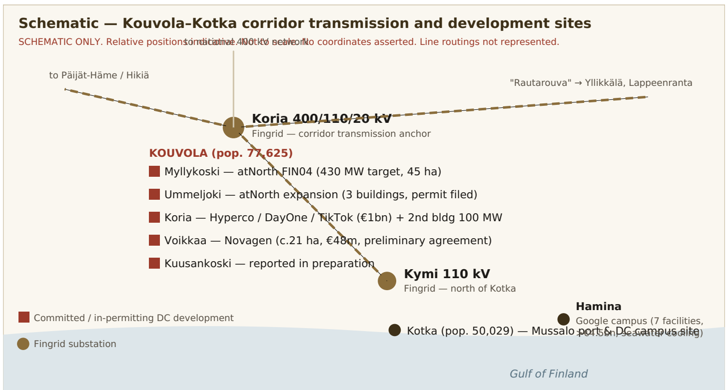


Figure 2. Corridor schematic. Note that the four Kouvola development clusters sit within a short radius of the Koria 400 kV anchor — the same electrical catchment Atlas would need to draw from.

7. Available Capacity

This is the most important investigation in the dossier, and it terminates in a negative result.

AVAILABLE MW NOT VERIFIED

Atlas has found no public source that quantifies uncommitted demand-connection capacity at any node in the Kouvola–Kotka corridor. No capacity figure appears anywhere in this dossier as a corridor fact.

What Atlas searched for, and what it found

Sought	Result	Detail
Node-level connection capacity, Koria / Kymi	UNKNOWN	Not published. Fingrid operates a "Grid Scope" map service showing grid connectivity for generation, consumption and — newly — regional connection capacity for grid energy storage. [FI02-GRD-015] Atlas has not extracted node-level demand headroom for this corridor from it and does not assert any.
Connection queue position / disclosure	UNKNOWN	Fingrid does not publish a queue. It has moved to collect standardised data on large consumption and storage connections planned via DSOs and directly. [FI02-GRD-016]
Restriction status for large demand, southern Finland	BLOCKER	See below — this is the finding that governs.
Aggregate national connection interest	CONFIRMED	Fingrid disclosed enquiries totalling around 500,000 MW of production and consumption connection interest against a peak consumption of roughly 15,000 MW. [FI02-GRD-008] Separately, more than 30,000 MW of grid energy storage connection requests. [FI02-PWR-015]
Reinforcement that would relieve the corridor	INDICATED	Hikiä–Toivila (2029) is cited as increasing connection capacity in southern Finland. [FI02-GRD-013] Whether it relieves Kymenlaakso specifically is not established.

The restriction, stated precisely

In January 2025 Fingrid announced local restrictions on connecting new industrial electricity consumption facilities to the main grid in southern Finland **for the period 2025 to 2027**, after which reinforcement investments complete. Fingrid described the practical effect as large new consumption projects potentially waiting longer than normal to connect, with the limitation applying to new industrial-scale demand facilities that had not yet agreed a connection. [FI02-GRD-006] CONFIRMED

Fingrid's Unit Manager for Main Grid Services was reported stating that **no new customers on the west coast or in southern Finland can be connected to the main grid until 2027 at the earliest.** [FI02-GRD-007] CONFIRMED

A related approval procedure now applies: approval must be sought from Fingrid for a connection contract for new consumption or grid storage, or for increasing an existing contract's capacity. Initially this applies to consumption connections above 2–10 MW and storage connections above 1–5 MW in southern Finland, with lower limits varying by region. [FI02-GRD-016]

The restriction on *grid energy storage* in southern Finland was subsequently extended to 2029, tied to Hikiä–Toivila completion. [FI02-GRD-013] Atlas notes carefully that this extension applied to storage, not to consumption — but it is direct evidence that southern Finnish connection constraints have proven **more durable than initially announced**, and that Fingrid is willing to extend restrictions when reinforcement slips.

> **Contradiction disclosed — C1**
>
> A DC Byte assessment quoted in trade press characterises Fingrid as "largely accommodating of data center projects, often prioritizing grid connections for operators and offering positive usage guarantees." [FI02-GRD-017] This is difficult to reconcile with Fingrid's own January 2025 restriction announcement and with the reported statement that no new southern Finland customers can connect before 2027. Atlas resolves the contradiction in favour of the primary source (Fingrid) over the Tier-3 source (a consultancy assessment quoted in secondary press), and notes the resolution rule: **a Tier-3 characterisation cannot override a Tier-1 announcement.** The DC Byte statement may be accurate for northern Finland, where Fingrid has stated that connections can still be implemented on a fast schedule by international standards. [FI02-GRD-018]

Why the corridor's existing projects do not prove capacity is available

A naive reading of Table 1 concludes that because six projects are proceeding, capacity must exist. This is the prohibited inference **infrastructure activity → infrastructure availability**, and it is wrong for three reasons.

1. **The restriction applies to facilities that had not yet agreed a connection.** Projects that secured connection agreements before the restriction are unaffected. atNorth's FIN04 was announced December 2023 and broke ground in 2024; the Hyperco Koria substation contract was let in 2025. These are grandfathered positions, not evidence of current headroom.
2. **The projects consume the headroom.** If atNorth's 430 MW target and Hyperco's two Koria buildings are delivered, they represent a large multiple of any plausible residual capacity at Koria. Each committed megawatt reduces what remains.
3. **Fingrid contracts against permits, not intent.** Fingrid's procedure requires a legally valid zoning plan and building permit before a connection agreement can be made, and Fingrid has stated it will in principle only agree grid connection capacity once customer permits are legally valid. [FI02-GRD-019, -020] Atlas holds no land, therefore no zoning, therefore no permit, therefore no possibility of a connection agreement. **The sequencing alone forecloses Atlas.**

8. Large-Load Connection Process

Atlas reconstructs the following pathway for a 100–500 MW demand facility in Finland. Timings are given only where authoritative evidence exists; where it does not, the cell says so.

Stage	Step	Counterparty	Indicative timing	Notes and evidence
1	Initial enquiry / connection point identification	Fingrid; relevant DSO	Not established	Fingrid's Grid Scope map service is the published entry point for identifying suitable connection points. [FI02-GRD-015]
2	Land control and zoning	Municipality; landowner	Zoning amendment "often lengthy, potentially taking years and involving multiple rounds of appeals"	Where the plan does not already permit data centre use, an amendment is required. Projects are typically sited near substations on undeveloped land or former industrial sites where zoning already permits the use. [FI02-REG-008] CONFIRMED
3	Building permit	Municipal technical committee	Case-specific	Kouvola's Technical Committee considered Hyperco's second Koria permit in June 2026, weeks after filing; the developer requested permission to begin work before the permit came into effect so foundations could be laid before ground freeze. [FI02-LND-006] This suggests municipal processing can be fast *where zoning is already correct*.
4	Connection agreement approval	Fingrid	Gated on stages 2–3	Fingrid requires a legally valid zoning plan and building permit before a connection agreement can be made. [FI02-GRD-019] A conditional connection agreement instrument has been introduced. [FI02-GRD-021]
5	Technical studies and compliance	Fingrid	Ongoing through 2026–27	See below — the requirements changed materially in June 2026.
6	Customer substation and cabling	EPC contractor	c.12–18 months observed	Eltel's Koria scope — turnkey 110 kV substation plus cabling to the Fingrid substation — ran from summer 2025 to planned completion in 2026 at c.€16m. [FI02-GRD-009] This is a usable real-world benchmark for a single connection at this scale.
7	Network reinforcement (if required)	Fingrid	New 400 kV line: 7–8 years planning to energisation	Reported figure. [FI02-GRD-022] INDICATED — single secondary source; Atlas has not corroborated against a Fingrid primary statement.
8	Construction and energisation	EPC; Fingrid	c.2 years observed for the building	Hyperco Koria: ground broken end-October 2025, completion May 2027. atNorth Myllykoski building: YIT contract July 2026, completion end-2027. [FI02-MKT-004, -006]

Table 4. Reconstructed large-load connection pathway. Stage 4 is the choke point: it is gated on land Atlas does not hold.

The June 2026 technical requirements — a new and material cost line

On **12 June 2026** Fingrid published a draft of KJV2026, new Grid Code Specifications for Demand Connections, opening a public consultation running to **21 August 2026**. On the same day Fingrid released a separate document, *Technical requirements for large demand facilities*, which applies **immediately** to demand facilities connecting directly to the transmission grid. [FI02-GRD-023] CONFIRMED

The requirements bite at **≥30 MW** and cover data centres, electrolysers and electric boilers. Large demand facilities are required, among other things, to **withstand grid disturbances** — a fault ride-through requirement for demand — and to have the **capability to limit their power intake** from the system. [FI02-GRD-024] They apply to new facilities and to modifications of existing ones, including power upgrades; they are not retroactive to unchanged existing plant. Fingrid targets Energy Authority confirmation of KJV2026 in early-to-mid 2027, having planned submission in October–November 2026. [FI02-GRD-023, -025]

Why this matters commercially. Fault ride-through and intake limitation are not paperwork. They dictate UPS topology, transformer specification, protection schemes, control systems and dynamic modelling deliverables. Any Atlas financial model built on pre-2026 Finnish connection assumptions is stale. Fingrid may additionally set site-specific study requirements at individual connection points — subsynchronous interaction, converter interactions, power quality. [FI02-GRD-026] Atlas has not costed compliance and cannot.

> **Timing note.** The KJV2026 consultation closes on 21 August 2026 — ten days after this dossier's date. Any party with a Finnish position should be filing comments. Atlas has no position and no standing to comment, which is itself a small illustration of the candidate's core problem.

9. Power Scenarios

Atlas models four scales. All four share a single fatal precondition — a connection agreement Atlas cannot obtain — so these scenarios describe what *would* be required rather than what is available. They are retained because they are portable to other candidates.

Parameter	A — 50 MW	B — 100 MW	C — 250 MW	D — 500 MW+
Framing	Initial AI deployment; single hall	Initial hyperscale campus	Large AI campus	Strategic European AI infrastructure campus
Connection architecture	Likely 110 kV customer substation off the regional network	110 kV, potentially dual-fed; direct Fingrid connectee	400/110 kV customer substation; direct transmission connection	Dedicated 400 kV bay(s); probable Fingrid substation extension or new station
Voltage	110 kV	110 kV	400 kV in / 110 kV distribution on site	400 kV
Transformer requirement	c.2 × 63 MVA class	c.2–3 × 63–100 MVA	Multiple 100–200 MVA units	400 MVA class units — the same class Fingrid is procuring from Hitachi Energy with multi-year lead times [FI02-GRD-012]
Redundancy	N+1 on site; single grid feed likely acceptable	N+1; dual feed strongly preferred	Dual independent feeds required for hyperscale tenancy	Dual 400 kV feeds from separate corridors; system-level stability studies mandatory
Reinforcement likely?	Possibly none	Possible	Probable	Near-certain — and on a new-400 kV-line timescale reported at 7–8 years [FI02-GRD-022]
Land requirement	c.5–8 ha	c.10–15 ha	c.25–35 ha	c.45–60 ha — benchmarked to atNorth's stated 430 MW across 45 ha [FI02-MKT-004]
KJV2026 applicability	Yes (>30 MW)	Yes	Yes	Yes; site-specific studies probable
Indicative build timeline	c.24 months from permit	c.24–30 months	Phased, 4–6 years	Phased, 6–10 years including reinforcement
Principal risk	Connection restriction window	Connection restriction; tax base	Reinforcement dependency; transformer lead time	Reinforcement timeline exceeds any plausible Atlas funding runway
Evidence confidence	UNVERIFIED	UNVERIFIED	UNVERIFIED	UNVERIFIED

Table 5. Power scenario comparison. Every capacity, timeline and equipment figure here is an Atlas engineering inference, not a Fingrid statement. See Appendix I for assumptions.

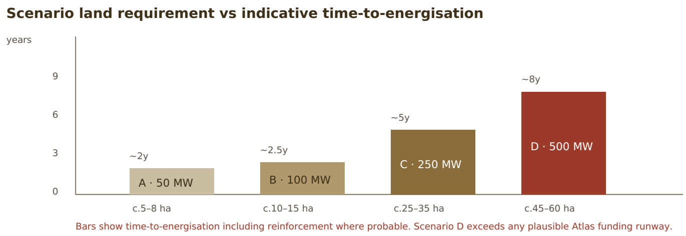


Figure 3. The scale Atlas would need to be strategically relevant (C–D) is precisely the scale at which reinforcement dependency dominates the timeline.

10. Electricity Economics

The most common error in data centre siting material is to present the wholesale price as the power cost. It is typically less than half of it. Atlas constructs the stack below as an **illustrative model with explicitly labelled assumptions** — it is not a quotation, not a Fingrid tariff extract, and not a basis for underwriting.

Delivered power cost stack — illustrative

Component	Illustrative €/MWh	Basis	Notes
Wholesale energy (day-ahead average)	41	STRONGLY INDICATED	Finnish 2025 annual average. [FI02-PWR-011] A 24/7 flat load captures approximately the time-weighted average; a flexible load can beat it, a peak-coincident load cannot.
Electricity tax — category I	**22.4**	CONFIRMED	**2.24 c/kWh from 1 July 2026.** Previously 0.05 c/kWh (€0.50/MWh) under category II. [FI02-REG-001]
Strategic stockpile fee	0.85	CONFIRMED	0.085 c/kWh; increased 1 April 2026. [FI02-REG-002]
Transmission charge (Fingrid)	Not established	UNKNOWN	Atlas has not extracted the applicable Fingrid tariff. Must be obtained before any model is built.
DSO charge (if applicable)	Not established	UNKNOWN	Not applicable if connected directly to Fingrid.
Balancing / imbalance cost	Not established	UNKNOWN	Material for a load with fast ramp characteristics — precisely the AI training profile.
Guarantees of origin	Not established	UNKNOWN	Small relative to the stack in a grid already at c.32 gCO2/kWh.
Identified subtotal		**c.€64/MWh — with three cost lines still unquantified**	

Table 6. Illustrative delivered power cost stack. Three components remain UNKNOWN; the true delivered cost is above the subtotal, not at it. The wholesale price is 64% of the identified subtotal and a smaller fraction of the true total.

The single most consequential number in this dossier

Before 1 July 2026 the electricity tax line was **€0.50/MWh**. After it, **€22.40/MWh**. On the illustrative stack above, tax moved from 1.2% of identified delivered cost to **35%**.

For a 100 MW campus operating at high load factor — roughly 850 GWh/year — the delta of 2.19 c/kWh is on the order of **€19 million per year** in additional electricity tax, gross of any relief. The Government's own estimate of the reform's yield is €47m annually at 2026 levels across the whole sector [FI02-REG-001], which implies that a handful of large campuses account for most of it. *Atlas assumption: 850 GWh/year at 100 MW. Actual consumption depends on load factor and PUE and is not established.*

PUE and the multiplier that is often forgotten

Google reports a PUE of approximately 1.1 at Hamina. [FI02-WTR-002] In a cold-climate free-cooling location, overhead is genuinely low — but the electricity tax applies to **total facility consumption**, not IT load. A 100 MW IT load at PUE 1.1 draws 110 MW at the meter. Every efficiency point is now worth 2.24 c/kWh more than it was in June 2026, which is the intended behavioural effect of the reform and is consistent with the state aid scheme's efficiency conditionality.

11. Power Procurement

Atlas identifies the following procurement routes and Finnish counterparties. **No approach has been made to any of them and no availability is asserted.** The prohibited inference *renewable generation → obtainable PPA* applies throughout.

Route	Corridor relevance	Assessment
Corporate wind PPA	High	The mainstream Finnish route. Google's Hamina campus has been supplied via wind farms since the original build, and Google reported achieving 98% carbon-free energy in Finland in 2023 with over three-quarters matched by contracted CFE within the Finnish grid region. [FI02-PWR-016] Wind resource is national, not corridor-specific — no local advantage.
Nuclear	Moderate	Loviisa sits roughly 60 km west of Kotka; Olkiluoto is on the west coast. Finnish nuclear is largely contracted through the Mankala cost-price model, which is a structural feature that a new entrant cannot easily access. UNVERIFIED
Hydro	Low	Concentrated in the north (Kemi, Oulu river systems). The Kymijoki has historic run-of-river hydro but at a scale irrelevant to a hyperscale campus.
Solar	Low	Growing fast — 402 MW of solar connected nationally in H1 2026, exceeding wind's 108 MW in the same period [FI02-PWR-008] — but the seasonal profile is poorly matched to a northern winter-peaking system.
Guarantees of origin	High	Straightforward and cheap in a 32 gCO2/kWh grid. Also close to meaningless as a differentiator, since every competitor has the same access.
24/7 CFE matching	High	Genuinely achievable in Finland to an unusual degree, given nuclear baseload plus wind. This is a real national selling point — and again, not a corridor one.
Behind-the-meter generation	Low	No identified brownfield generation asset in the corridor available for repurposing at the required scale.
BESS / microgrid	CONSTRAINED	**Materially restricted.** No new large-scale grid storage (>5 MW) may connect in southern Finland before 2029, pending Hikiä–Toivila. All storage projects above 5 MW must submit a connection contract enquiry to Fingrid. [FI02-GRD-013] This closes off the obvious mitigation for a connection-constrained site.

Table 7. Power procurement routes. Note the compounding failure: the connection constraint blocks the campus, and the storage constraint blocks the workaround.

The regulatory direction of travel

A legislative proposal advanced by the Kokoomus (National Coalition) party would, if enacted, **require large data centres to obtain matching new power generation capacity before connecting to the Fingrid-operated national grid.** The proposal was reported as supported by the Social Democrats, Finland's main opposition party. [FI02-REG-007] INDICATED — this is a legislative proposal, not law, and Atlas has not verified its current parliamentary status.

Atlas flags this as the most important forward-looking regulatory risk in the power section. If enacted, it converts a data centre developer into a generation developer — a fundamentally different business with a different capital structure, different timescales and different skills. It would raise the barrier to entry sharply, to the further advantage of incumbents already holding connection agreements. It would also, in fairness, be a coherent policy response: Finnish Energy's own former Fingrid CEO publicly welcomed removal of the tax concession on the grounds that there was no justification for the lower rate. [FI02-REG-009]

Part II conclusion

The Finnish power system is genuinely excellent for AI infrastructure: clean, cheap, reliable, single-zone. **None of that is corridor-specific.** What is corridor-specific — node capacity — is unknown, sits inside an administratively restricted region, and is being consumed by six committed competitors. The fiscal advantage that made the national case compelling was withdrawn on 1 July 2026, and the obvious technical mitigation (storage) is itself restricted until 2029. Power does not support this candidate.

PART III

Land & Physical Site

12. Regional Landscape

Kymenlaakso is a region of roughly 4,559 km² on the Gulf of Finland, with Kotka and Kouvola as twin regional capitals and Hamina as the third town. Its economy was built on the Kymijoki river: pulp, paper, sawmilling, and the port trade that carried the output. That base has been contracting for three decades, and the demographic record is unambiguous.

Indicator	Value	Source and note
Kymenlaakso population, 1990	186,030	Statistics Finland series. [FI02-WRK-001]
Kymenlaakso population, 2010	175,377	Same series.
Kymenlaakso population, 2020	162,812	Same series.
Kymenlaakso population, 2025	**156,198**	**A 16% decline over 35 years, and 4% in the last five.** [FI02-WRK-001] CONFIRMED
Kouvola (city), 31 Dec 2025	77,625	12th largest Finnish municipality; urban area c.55,372; city centre urban area c.47,000. [FI02-WRK-002]
Kotka, 31 Dec 2025	50,029	[FI02-WRK-003]
Hamina, 31 Dec 2025	19,113	Central town c.10,000. [FI02-WRK-004]
Kotka–Hamina sub-region total	78,573	Includes Miehikkälä 1,664; Pyhtää 5,016; Virolahti 2,751. [FI02-WRK-003]
Regional GDP per capita (2015)	€34,350	Total €6.158bn. Dated but the most recent Atlas located. [FI02-WRK-005] INDICATED
Distance Kouvola → Helsinki	c.134 km	[FI02-LOG-005]
Distance Kouvola → Hamina	c.52 km	[FI02-MKT-002]
Distance Kotka → Helsinki region	c.130 km	[FI02-WRK-006]

Table 8. Regional indicators. The population trend is the single most important workforce datum in this dossier and is addressed in §31.

Structural change and the just-transition thesis

The industrial contraction that created this corridor's brownfield estate is documented and recent. In June 2023 Stora Enso decided to close the Sunila pulp mill in Kotka and to close one of two paper machines at its Anjala production unit, as part of a wider restructuring affecting approximately 1,150 employees group-wide with around 600 job losses at the affected facilities. [FI02-MKT-013] CONFIRMED

The Finnish Government responded with dedicated structural-change support, strengthening employment and business services in Kymenlaakso by €1 million, and explicitly linked the region's difficulties to the closure of the Russian border and its effects on forest industry, logistics and tourism. [FI02-MKT-014] CONFIRMED

This is the classic Atlas target profile: heavy electrical infrastructure and large levelled sites left behind by departing industry, in a region with political motivation to attract replacement investment. The Voikkaa estate is the archetype — a UPM paper mill that ceased operations in 2006, since redeveloped to host around 30 smaller companies. [FI02-LND-004] Google's Hamina campus is the proof of concept, built on a Stora Enso mill closed in January 2008 and sold for approximately €40 million. [FI02-MKT-001]

> **The methodological lesson, stated plainly.** Atlas's just-transition site-selection thesis is **correct**. This corridor validates it. The problem is that the thesis is not proprietary: atNorth, Hyperco, Novagen and Google all executed it here, and Redeve — a real estate investment firm that acquired both the Myllykoski and Voikkaa mill estates — built a business on it. A methodology that is right and widely known produces no excess return. Atlas should extract the *sequence* from this candidate and apply it earlier in the cycle elsewhere.

13. Actual Site Search

Atlas identifies the following candidate site areas from public sources. The critical column is the last one.

Site	Municipality	Hectares	Ownership	Zoning	Power proximity	Water	Status — availability to Atlas
Myllykoski mill estate	Kouvola	21 ha sold; >50 ha retained	Redeve (retained portion)	Industrial	Koria catchment	Kymijoki	**Partially available.** Redeve sold 21 ha to atNorth and states it retains more than 50 ha in the industrial park which it continues to develop. [FI02-LND-001] STRONGLY INDICATED This is the *only* identified large parcel in the corridor plausibly open to a third party — and Atlas would be bidding beside atNorth's expanding campus, against atNorth.
Ummeljoki / Myllykoski expansion	Kouvola	Part of 45 ha campus	atNorth	Industrial	Adjacent to FIN04	Kymijoki	**Not available.** Permit filed 2026 for three additional data centre buildings. [FI02-LND-002] TAKEN
Koria industrial plot	Kouvola	24 ha	Hyperco (bought from City of Kouvola, 2024)	Industrial	Adjacent to Fingrid Koria substation	Kymijoki	**Not available.** Under construction; second building in permitting. [FI02-LND-003] TAKEN
Voikkaa mill	Kouvola	c.20–21 ha	Redeve → Novagen	Industrial	Koria catchment	Kymijoki	**Contracted.** Preliminary agreement

estate			Data Centre Oy				signed 20 February 2026; €48m; €200,000 deposit paid. [FI02-LND-004] EFFECTIVELY TAKEN
Kuusankoski	Kouvola	Not established	Not established	Not established	Koria catchment	Kymijoki	**In preparation by others.** Yle reported March 2026 that data centres are under construction or in preparation at Kuusankoski alongside Myllykoski, Ummeljoki and Koria. [FI02-LND-005] STRONGLY INDICATED
Kouvola Savontalo DC site	Kouvola	10 ha; 22,800 m² building rights	Marketed via datacenterfinland.fi	**Social/educational services** — re-zoning required	Near DSO substation at Nirvistentie; Koria to the south	District heating present	**Marketed, but flawed.** Vendor claims 30 MW short-term and >100 MW long-term, "excellent soil conditions", possibility to extend to nearby plots, existing district heating with a disused backup heat plant on site. Vendor states discussions with the municipality indicate re-zoning could be executed within a year. [FI02-LND-007] UNVERIFIED – VENDOR CLAIM See warning below.
Kuusaa DC	Kouvola	Not established	Not established	Not established	"Substantial power grid capacity in the region" (vendor)	Not established	**Listed in the Business Finland data centre site catalogue.** [FI02-LND-008] UNVERIFIED
Kotka Mussalo DC Campus	Kotka	26 ha	City of Kotka	Pre-zoned per regional marketing	Not established	Seawater; district heating to Kotka centre adjacent	**Listed as reserved until 04/2024** in the Business Finland catalogue — current status unknown. Solid rock/moraine; plot to be levelled by the owner; 3 km from Kotka centre at Mussalo Harbour; 6 km to E18; 130 km to Helsinki Airport. [FI02-LND-009] STATUS UNKNOWN
Google Hamina land bank	Hamina	50 ha	Alphabet / Tuike Finland Oy	Industrial	Existing campus infrastructure	Seawater	**Not available.** Acquired 2022 for €4.3m adjacent to the existing campus explicitly to accommodate possible expansion. [FI02-LND-010] TAKEN
Ahvenkoski DC Site	Pyhtää	Not established	Not established	Not established	Not established	Coastal	**Listed as reserved until 09/2024** in the Business Finland catalogue. Just outside the corridor proper. [FI02-LND-011] STATUS UNKNOWN
Loviisa (adjacent)	Loviisa	c.6 ha phase 1	City of Loviisa → Nova Complex Oy (lease)	Industrial	Near E18 eastern exit	Coastal	**Not available.** Land lease and preliminary agreement approved by City Council 21 January 2026; three-phase complex planned; planning reservations for surrounding land. [FI02-LND-012] TAKEN

Table 9. Candidate land register. Of eleven identified areas, four are confirmed taken, one is contracted, one is in preparation by others, two carry expired reservation flags with unknown current status, and two are unverified vendor listings. ***One — the Redeve retained estate at Myllykoski — is plausibly open.***

Warning: the Savontalo listing commits three prohibited inferences

The marketed Kouvola Savontalo site illustrates exactly what Atlas methodology exists to catch. It advertises ">100 MW long term" — **transmission proximity presented as available MW**, with no Fingrid allocation cited. It advertises the site's proximity to the Koria hub and asserts "exceptional power infrastructure" because of previous forest industry — **historical industrial load presented as current headroom**, ignoring that the load was decommissioned decades ago and the capacity reallocated. And it presents a re-zoning from social/educational services as executable "within a year" based on **discussions with the municipality** — municipal enthusiasm presented as permitting certainty, against a specialist legal assessment that Finnish zoning amendments are "often lengthy, potentially taking years and involving multiple rounds of appeals". [FI02-REG-008] The listing also appears to date from an earlier market cycle. Atlas assigns it UNVERIFIED and would not act on any figure in it.

14. Land Control

Atlas distinguishes five states of land availability. The corridor distribution is the finding.

State	Count	Corridor detail
Advertised	3	Savontalo, Kuusaa, Kotka Mussalo — all via the Business Finland data centre site catalogue or an associated marketing site. Two carry reservation flags dated 2024. Advertised status tells Atlas nothing about current availability.
Municipally owned	1–2	Kotka Mussalo (City of Kotka). The Koria plot *was* municipal and was sold to Hyperco in 2024 — an instructive precedent for how quickly municipal land moves in this market.
Privately owned	4+	Redeve (Myllykoski retained, Voikkaa), atNorth, Hyperco, Alphabet. The two large convertible mill estates were both held by a single specialist owner, Redeve, which has now monetised both.
Optionable by Atlas	**0 identified**	Atlas has identified no parcel over which it could realistically obtain an exclusive option. The Redeve retained estate is the only theoretical candidate, and Redeve's demonstrated counterparties are institutionally-backed developers paying €48m for 21 ha.
Constrained	1	Savontalo — zoned for social/educational services, requiring an amendment before any data centre use.

Finnish land acquisition mechanisms and the defence permit

Two features of the Finnish regime are material to Atlas.

Municipal land leasing is a live route. The Loviisa transaction is the clearest recent template: the City Council approved a land lease agreement plus preliminary agreement, leasing approximately six hectares for phase one, built on a plot reservation granted in autumn 2025 that gave the developer the right to apply for permits and begin preconstruction. [FI02-LND-012] The plot-reservation instrument is how a developer without capital secures optionality — and it is granted by municipalities to counterparties they judge credible.

Non-EU/EEA acquirers need Ministry of Defence permission. For real estate transactions, non-EU and non-EEA purchasers require a permit from the Ministry of Defence to purchase real estate in Finland where their ownership or equivalent effective influence exceeds 10%. [FI02-REG-010] CONFIRMED This is not a barrier for Albedo Industries as a Spanish entity, but it is a live consideration for the capital Atlas would need to attract, and it is a live consideration for at least one active corridor participant, given Hyperco's ownership by Dubai-based Damac's Edgnex.

Security scrutiny is increasing. The Ministerial Committee on Economic Policy decided in December 2025 that security authorities would explore security aspects related to data centres and must have sufficient access to information on data centre investments and operations, alongside a new registration obligation. [FI02-REG-003]

Conclusion on land control

Atlas cannot realistically establish site control in this corridor. The convertible estate is held by a small number of owners who have demonstrated they transact with institutionally-backed developers at eight-figure prices. Atlas holds no capital commitment, no track record in this asset class, and no relationship with Redeve, the City of Kouvola or the City of Kotka. A municipal plot reservation is theoretically obtainable but is granted on developer credibility, which Atlas would have to build before it could ask. This is a market-access failure, not an information failure, and no additional research resolves it.

15. Physical Conditions

Atlas has established relatively little at plot level and says so. What follows is a mixture of confirmed regional characteristics and unverified site-specific vendor claims.

Factor	Class	Finding
Seismicity	CONFIRMED	Southern Finland is a very low-seismicity area. Minor local events have occurred in the Kotka–Hamina area but are small and short in duration and are not required to be accounted for in Finnish building regulations, per the University of Helsinki Institute of Seismology as cited in the Kotka Mussalo site documentation. [FI02-LND-013] **A genuine and material advantage over several competing European geographies.**
Geology — Kotka Mussalo	INDICATED	Solid rock / moraine; the plot would be levelled by the owner (City of Kotka). [FI02-LND-009] Rock is excellent for foundation loading but raises excavation cost for deep services.
Geology — Kouvola inland sites	UNVERIFIED	Vendor material claims "excellent soil conditions" and a "mostly levelled plot" at Savontalo. [FI02-LND-007] No geotechnical report seen. Note that the mill estates carry a genuine advantage here: heavy industrial buildings imply foundations already proven for heavy loads.
Flood risk — Kotka Mussalo	INDICATED	Site material cites the Finnish Meteorological Institute's June 2014 study on long-term flooding risks and minimum building elevations on the Finnish coast: minimum recommended building elevation without wave compensation at Mussalo is 314 cm above sea level (interpolated from Helsinki and Hamina mareograph data, based on 2100 sea level at a 1-in-250-year exceedance). The estimated minimum plot level is stated as +500 cm. [FI02-LND-014] **This is unusually rigorous site documentation and should be treated as a positive signal about Kotka's site-preparation competence.**
Flood risk — inland	UNKNOWN	Kymijoki floodplain exposure at specific Kouvola plots not established.
Contamination	UNKNOWN	**Material gap.** Every large candidate parcel in Kouvola is a former pulp or paper mill. Historic mill sites routinely carry soil and groundwater contamination. No environmental site assessment has been seen for any parcel. Remediation liability could be a multi-million-euro line item and is entirely unquantified. **This would be a Gate 2 priority on any candidate that survived Gate 1.**
Previous industrial use	CONFIRMED	Voikkaa: UPM paper mill, ceased 2006, since hosting c.30 companies. Myllykoski: paper mill estate held by Redeve. Hamina Summa: Stora Enso mill, closed January 2008 after 53 years. Sunila, Kotka: Ahlström-origin pulp mill from 1938, closed 2023. [FI02-LND-004, -001; FI02-MKT-001, -013]
Heritage constraint	INDICATED	**Specific to Sunila.** The Sunila industrial and residential area in Kotka was designed by Alvar Aalto and is documented by DOCOMOMO Finland as a heritage ensemble. [FI02-LND-015] Any redevelopment of the Sunila mill site would face heritage scrutiny that does not apply to Voikkaa or Myllykoski. Atlas flags this because Sunila is the most obvious "recently closed pulp mill with grid connection" in Kotka and would be the first place a naive screen would look.
Terrain and drainage	UNKNOWN	Not established at plot level for any candidate.

Table 10. Physical conditions. The contamination gap and the Sunila heritage constraint are the two findings a purely desk-based screen would most easily miss.

Connectivity

16. Terrestrial Fibre

Connectivity is the weakest evidence area in this dossier and Atlas states that plainly rather than filling the gap with inference.

Carriers present in the Finnish market

Carrier	Ownership	Relevance to the corridor
Cinia Oy	Majority state-owned (reported at more than 77%)	Operates a fibre network reported at approximately 17,000 km spanning Finland's largest cities, **strategically positioned along railways** for direct city-to-city connections, using ITU-T G.652.D compliant cable. Owns and operates C-Lion1. [FI02-FBR-001, -002] CONFIRMED
Telia Finland	Telia Company	Fibre, 4G and 5G networks covering 99% of Finland's population; secured backbone from Helsinki to Utsjoki; regional networks reaching small localities. States data centre connectivity can be built using four separate fibre routes and services from multiple operators. [FI02-FBR-003, -004]
Elisa Oyj	Listed	One of the three national operators; also builds edge data centres in Finland. [FI02-MKT-016]
DNA Oyj	Telenor	Third national operator. Migrating 100 core applications to Google Distributed Cloud for data residency — a demand-side signal for Finnish capacity. [FI02-MKT-017]
GlobalConnect, Arelion, Lumen, RETN, others	Various	Present in the Finnish wholesale market per Telia's published carrier list. [FI02-FBR-004] Corridor presence not established.
Lounea	Finnish	Regional operator that appointed a Kouvola-based executive to lead fibre network business in Eastern Finland. [FI02-FBR-005] INDICATED — dated source; current relevance unverified.

The Kouvola railway junction inference — and its limits

Cinia states that its fibre network is positioned along railways. [FI02-FBR-002] Kouvola is one of Finland's busiest railway crossing points, with traffic in four directions, roughly 130 km from Helsinki Central. [FI02-LOG-004] It is therefore **highly likely** that Cinia long-haul fibre passes through or near Kouvola.

> **Atlas does not convert this into a capability claim.** "Backbone fibre passes nearby" and "two genuinely diverse, hyperscale-grade, high-capacity routes are procurable to a specific plot at a known price and lead time" are different statements separated by significant engineering and commercial work. This is the prohibited inference **fibre presence → hyperscale redundant connectivity**. Atlas classifies corridor fibre as INDICATED and node-level route diversity as UNKNOWN.

17. International Connectivity

C-Lion1 — and the finding that matters

C-Lion1 is the first direct submarine communications cable between Finland and Central Europe, owned and operated by Cinia, running approximately 1,173–1,175 km to Rostock, Germany. Before it, Finnish international traffic routed via Sweden and Denmark. It comprises eight optical fibre pairs with a reported transmission capacity of 144 Tbit/s. Total project cost was approximately €100 million, with the Finnish Government an enabler and investor alongside institutional investors. [FI02-FBR-006, -007] CONFIRMED

> **Key finding — the corridor is not a cable landing geography**
>
> **C-Lion1's Finnish landing points are Helsinki and Hanko — not Kotka, not Hamina, not Kouvola.** [FI02-FBR-006] A network switch was installed at Hanko in October 2017. Any pitch positioning this corridor as "adjacent to Finland's subsea gateway to continental Europe" is geographically incorrect. Traffic from Kouvola or Kotka to Rostock traverses terrestrial fibre to Helsinki or Hanko first. This does not disqualify the corridor — Google has operated at Hamina for fifteen years — but it removes a claimed advantage and adds a terrestrial route dependency that must be engineered and priced.

Route diversity and physical security

The Baltic subsea environment carries an elevated and demonstrated risk profile. C-Lion1 suffered a fault on 18 November 2024, detected at 04:04, in the Swedish exclusive economic zone east of Öland. Cinia's CEO stated the cable had never broken before; a company spokesperson said cables do not break without external impact and noted no seismic events could account for it. Germany's Federal Defence Minister characterised the incident as sabotage. Repair completed ahead of schedule on 28 November 2024, using a vessel dispatched from France. [FI02-FBR-008, -009] CONFIRMED

Finland's National Cyber Security Centre noted that connections out of Finland run from several different places. [FI02-FBR-008] This is accurate and reassuring at national level. It is **not** evidence of diverse routing to a specific plot in Kymenlaakso.

Route requirement	Class	Assessment
Two physically diverse terrestrial routes from corridor to Helsinki	UNKNOWN	Plausible given rail and motorway corridors, but unproven at plot level. Must be established through carrier engagement.
Onward diversity Helsinki → Continental Europe	INDICATED	C-Lion1 direct to Rostock, plus legacy routing via Sweden and Denmark. Genuine diversity exists at national level.
Latency to Frankfurt	UNVERIFIED	Cinia's marketing claim is that the route lowers latency to the point where Helsinki "appears as a suburb of Frankfurt". [FI02-FBR-010] Vendor claim; corridor adds terrestrial distance beyond Helsinki.
Baltic geopolitical exposure	RISK	Demonstrated. See risk R09 in §38. Any operator underwriting a Finnish campus must price a repeat of the November 2024 event.
Arctic Connect / eastern routes	UNVERIFIED	Cinia's Arctic Connect project to link Europe and Asia was reported as stalled. [FI02-FBR-011] The proximity of the closed Russian border removes any eastern transit thesis entirely.

Table 11. International connectivity assessment. Note the asymmetry: national-level diversity is good; plot-level diversity is unproven.

Cooling, Water & Heat

18. Climate & Cooling

This is the corridor's — and Finland's — least equivocal strength, and the one area where Atlas's opportunity score and evidence confidence are both high.

The engineering advantage

Southern Finland's climate permits extensive free cooling. Operators consistently and independently cite it: atNorth describes the cool climate as allowing efficient natural free air cooling; DayOne cites a naturally favourable climate that significantly reduces energy consumption and operational costs. [FI02-WTR-003, -004] Google reports a PUE of approximately 1.1 at Hamina using seawater from the Bay of Finland, drawn from the sea floor where temperature is more even, through pipes up to two metres in diameter, circulated using renovated twenty-year-old pumps from the original paper mill. [FI02-WTR-001, -002] CONFIRMED

High-density AI racks: where the advantage compounds and where it does not

The move to 100 kW and prospectively 200 kW+ racks changes the cooling calculus in ways that **favour** this geography, with two caveats Atlas states rather than glosses.

Consideration	Assessment
Air cooling at high density	A cold ambient extends the free-cooling window and reduces compressor hours, but air becomes volumetrically impractical well before 100 kW/rack regardless of climate. Climate advantage is real but bounded.
Direct liquid cooling	Where the advantage compounds. A low ambient permits higher approach temperatures in dry coolers, reducing or eliminating mechanical chilling for a warm-water DLC loop. atNorth's published FAQ material indicates DLC in its newer builds. [FI02-WTR-003] INDICATED
Waste heat quality	DLC raises return-water temperature, which materially improves district-heating off-take economics by reducing heat-pump lift. This is why the corridor's heat off-take arrangements (§20) are technically credible rather than greenwash.
Extreme cold	Requires freeze protection, glycol management, and careful humidity control. A design consideration, not an obstacle — fifteen years of Hamina operation settles the question.
Seawater cooling — Kotka/Hamina	Proven at scale. Available only to coastal sites, i.e. **not** to the Kouvola cluster where most available land sits. This is a genuine internal split within the "corridor".
Zero-freshwater designs	DayOne markets zero freshwater cooling in its Finnish campuses. [FI02-WTR-005] Increasingly the expected standard and reduces §19 exposure.

Honest assessment. Climate is a real advantage, but it is a *national* advantage shared with Sweden, Norway, Iceland and Denmark, and within Finland it is shared with every competing location. It is a reason to build in the Nordics. It is not a reason to build in Kymenlaakso rather than Lahti, Mäntsälä or Lappeenranta.

19. Water

Atlas has established little and assumes nothing. Water availability is **not** to be inferred from the presence of a river or a sea.

Source	Class	Assessment
Seawater — Gulf of Finland	CONFIRMED (Hamina only)	Google has cooled its Hamina campus entirely with Baltic seawater since 2011, reusing paper mill intake infrastructure. [FI02-WTR-001] This confirms technical and permitting feasibility *at that site with that legacy infrastructure*. It does not confer rights at any other site.
Seawater — Kotka Mussalo	UNVERIFIED	The site sits at Mussalo Harbour. Regional marketing refers to "bodies of water needed for cooling". [FI02-LND-016] No abstraction volume, intake route, temperature profile or discharge permit established.
Kymijoki — Kouvola cluster	UNKNOWN	The river that built the region's industry. Abstraction rights, seasonal low-flow constraints and thermal discharge limits are all unestablished. Historic mill abstraction rights may or may not transfer with land.
Municipal supply	UNKNOWN	No capacity figure obtained from any of the three municipal water undertakings.
Groundwater	UNKNOWN	Finnish groundwater areas carry protection designations that can constrain both abstraction and construction. Not checked at plot level. **A Gate 2 priority.**
Discharge / wastewater	UNKNOWN	Thermal discharge to the Gulf or the Kymijoki would require an environmental permit with a water-body impact assessment.

Table 12. Water assessment. Six rows; one confirmed, and only for a site Atlas cannot access.

20. Waste Heat

Waste heat is the corridor's most distinctive feature and — following the July 2026 reform — its most commercially important one, because **waste heat reuse is a stated condition of the replacement state aid scheme.** [FI02-REG-003] The corridor's three district-heating utilities have all engaged.

Utility	Municipality	Counterparty	Status
Haminan Energia	Hamina	Google	**Operational / commissioning.** Google's offsite heat recovery project with Haminan Energia and the municipality of Hamina. Recovered heat represents an estimated **80% of Hamina's district-heating demand**, serving households, schools and public service buildings in a region previously highly dependent on fossil fuel; expected to contribute to Hamina's energy balance from late 2025. Google states the Hamina campus operates at 97% carbon-free energy, so the recovered heat is 97% carbon-free. [FI02-HEA-001] CONFIRMED **This is one of the most significant data centre heat recovery projects in Europe.**
KSS Energia	Kouvola	atNorth (FIN04)	**Contracted / in design.** Excess heat from FIN04 to be recycled through the KSS heating plant to provide heating and hot water for the local community; KSS CEO Marko Riipinen publicly committed to the arrangement in December 2023 and it remains in the July 2026 YIT contract description. [FI02-HEA-002] CONFIRMED
Kouvola district heating	Kouvola	Hyperco (Koria)	**Planned.** The Koria development is described as integrating waste heat recovery contributing to the local district heating network. [FI02-HEA-003] STRONGLY INDICATED
Kotkan Energia	Kotka	None identified	Kotka operates a district heating network; the Mussalo site material notes the Kotka city centre heating network adjacent to the site offering "possibility for very cost efficient energy and heat re-use". [FI02-LND-009] **No data centre counterparty identified.** UNVERIFIED
Lahti Energy (adjacent)	Lahti	DayOne	Outside the corridor but instructive: DayOne's €1.2bn Lahti campus integrates waste-heat recovery with Lahti Energy. [FI02-HEA-004]

Table 13. Waste heat off-take landscape. Note the pattern: ***each utility has one data centre partner, and each partner is already chosen.***

The strategic implication Atlas must not miss

District heating off-take is a **scarce, exclusive-in-practice resource**. A district-heating network of a given size can absorb a finite heat load; once a network's demand is substantially met by one data centre — as at Hamina, at approximately 80% of demand — the marginal value of a second data centre's heat in that network falls close to zero. Kouvola's KSS network is committed to atNorth and Hyperco. Hamina's is committed to Google. **Kotka is the only corridor network without an identified partner**, which is the single most interesting residual fact in this dossier and the only reason §46 does not close the file outright. But Kotka's available land is a 26 ha municipal plot last publicly flagged as reserved in 2024, with unknown current status — and heat off-take without land and grid capacity is not a business.

Two further constraints on the heat thesis, both structural. First, **seasonality**: district heat demand in Finland is strongly winter-weighted, while a data centre rejects heat constantly. The off-take contract therefore monetises only part of the annual heat output unless seasonal storage or industrial off-take is available. Second, **temperature**: conventional Finnish district heating operates at supply temperatures well above data centre return-water temperatures, requiring heat pumps whose electricity consumption is itself now taxed at category I. The reform has, in this narrow respect, made heat recovery marginally harder to justify at the same time as making it a condition of aid.

21. Finland Data Centre Market Inventory

The Finnish market moved from a modest colocation base to a genuine hyperscale destination between 2023 and 2026. The inventory below is not exhaustive but captures every participant material to a Kymenlaakso assessment.

Operator	Location	Capacity	Investment	Status	Customer	Expansion / notes
Google	Hamina	Not disclosed	>€4.5bn cumulative; €1bn latest tranche	Operating; 7th facility announced	Own workloads / Google Cloud	Bought Summa mill from Stora Enso for c.€40m in 2009; first facility opened 2011; 6th opened 2020. 50 ha adjacent land bought 2022 for €4.3m. c.400–500 staff. Also acquired 1,400 ha from Metsähallitus in northern Finland. Heat recovery to Haminan Energia. [FI02-MKT-001, -008; FI02-LND-010]
atNorth	Kouvola (Myllykoski/Ummeljoki); Espoo; Vallila; others	FIN04: 430 MW target across 45 ha	YIT contract c. €300m for one building	Under construction; go-live reported H1 2027	Colocation + build-to-suit; "ongoing discussions with several AI companies, hyperscalers"	Owned by Partners Group since Dec 2021. Acquired FIN01 and FIN03 from Advania. Permit filed 2026 for three further buildings. Heat reuse via KSS Energia. Finland operations lead has publicly discussed recruitment: c.60 staff at phase one, c.200 later. [FI02-MKT-002, -004, -005; FI02-WRK-007]
Hyperco (Edgnex / Damac)	Koria, Kouvola; Espoo; Lohja; Pyhäjoki; Vantaa	Koria bldg 1: c.28,000 m² on 24 ha. Bldg 2: 100 MW, 56,670 m². Lohja: up to 300 MW	€1bn (Kouvola, per TikTok)	Bldg 1 under construction (completion May 2027); bldg 2 in permitting	**TikTok / ByteDance** confirmed	Founded 2020; acquired by Edgnex (Damac) April 2025. Prior investors included NREP and Varma. Acquired 130 ha in Pyhäjoki. Lohja land deal 217,420 m² approved. [FI02-MKT-003, -006, -007]
DayOne	Lahti (Kiveriö); Kouvola (with Hyperco)	Lahti: 128 MW IT total, 50 MW first building. Group Finland: 281 MW	c.€1.2bn (Lahti)	Lahti demolition from Q3 2025; RFS 2027	"Negotiations with potential tenants in progress, no agreements finalized" (Aug 2025)	Singapore-headquartered. Brookfield and a sovereign wealth investor provided a €1bn holdco financing. Lahti site 98,901 m², fully zoned industrial. Waste heat with Lahti Energy; zero freshwater cooling. [FI02-MKT-009, -010; FI02-FIN-001]
Novagen Data Centre Oy	Voikkaa, Kouvola	c.21 ha	€48m land	Preliminary agreement Feb 2026	Not disclosed	Established autumn 2025, registered office Kouvola. Chairman Tarun Tyagi also chairs Hyperco. Seller: Redeve. [FI02-LND-004]
Verne	Mäntsälä	Not established	Not established	Construction begun	Colocation / HPC	Established Nordic operator. [FI02-MKT-011]
Nebius / Polarnode	Lappeenranta (Pajarila)	Not established	Not established	Announced	AI cloud	AI-focused campus roughly 87 km east of Kouvola — the nearest direct AI-infrastructure competitor. [FI02-MKT-012] INDICATED
XTX Markets	Kajaani	Not established	>$1.1bn reported	**Under review**	Own quant workloads	Reported to be reviewing the investment following the electricity tax proposal; had signed a YIT contract for a second campus building. [FI02-REG-006]
Nova Complex	Loviisa	6 ha phase 1 of 3 phases	Not disclosed	Land lease approved Jan 2026	Not disclosed	Singapore-headquartered. Facilitated by Cursor Oy and the City of Loviisa. [FI02-LND-012]
FCDC	Rovaniemi (Hietavaara)	500 MW projected; c.180,000 m²	Not established	Construction targeted 2027; ops 2029	Not disclosed	Among the largest planned Finnish projects. [FI02-MKT-012] INDICATED — trade-press aggregation.
Others	Various	—	—	—	—	Equinix, Telia, Elisa, Hetzner, DataCrunch (Helsinki + proposed Akaa), Microsoft, QTS (Forssa), AmpTank (Utajärvi, 200 MW), Scale4 (Kitee), Arcem, Borealis, Aurinkokarhu (Närpiö, 60 MW), Bitzero (Kokemäki, 500 MW announced). [FI02-MKT-011, -015, -018]

Table 14. Finnish data centre landscape. Where a figure is not disclosed, the cell says so — Atlas does not estimate operator capacity.

Market size — the sources disagree, and Atlas discloses the disagreement

Contradiction disclosed — C2: Finnish installed capacity

Source	Installed	Pipeline / forecast	Tier
Confederation of Finnish Industries + Finnish Data Center Association, 2025 study	c.285 MW	>1.5 GW by 2030; announced projects >3.4 GW	Tier 1–2
Trade aggregator, May 2026	>700 MW	Pipeline >2 GW	Tier 3
Commercial database report, Mar 2026	24 existing facilities	41 upcoming; forecast >5 GW	Tier 3
Market research house, May 2026	Market value $952m (2025)	$5.8bn by 2031; 35.18% CAGR	Tier 3

Atlas resolution. The FDCA/industry figure of 285 MW installed rising past 1.5 GW by 2030 is the best-sourced and is adopted. The higher installed figures likely reflect different definitions of "capacity" (IT load vs facility power vs contracted). The 5 GW forecast is an outlier and is not used. **What all four agree on is direction and magnitude of growth — roughly a five-fold increase this decade.** [FI02-MKT-019, -020, -021, -022]

22. AI Infrastructure Investment, 2024–2026

Three things changed to produce the Finnish investment wave, and one of them has now reversed.

Driver	What happened	Status Aug 2026
1. Clean, cheap, abundant power	Olkiluoto-3 came online in 2023, adding 1,600 MW and pushing nuclear to around 40% of generation. Wind build-out continued. Finland became 98% self-sufficient in electricity production by 2023 with one of Europe's cleanest grids. [FI02-PWR-005, -009]	**Intact**
2. Direct connectivity to Central Europe	C-Lion1 opened in 2016, removing the Sweden/Denmark routing dependency and enabling the Nordics to be marketed as latency-viable for German and Central European workloads. [FI02-FBR-006]	**Intact** (but see §17 — not in this corridor)
3. A 45-fold electricity tax advantage	Data centres sat in category II at 0.05 c/kWh against a general rate of 2.24 c/kWh. The concession was worth an estimated €50m to the sector in 2025 and was explicitly credited with attracting Microsoft and Google. [FI02-REG-004, -009]	**REVERSED 1 July 2026**

The scale of the response was startling for a country of 5.5 million. Industry counted 22 new data hall projects in one year with at least 15 more in the following half-year. Fingrid faced connection applications reported at some 24,000 MW of new IT load — described as five times Finland's entire nuclear capacity — and municipalities from Helsinki to Oulu reported suitable industrial plots virtually sold out. [FI02-MKT-023] INDICATED — secondary source; Atlas notes the 24,000 MW figure is not independently corroborated and is smaller than, and therefore consistent with, Fingrid's own 500,000 MW aggregate enquiry disclosure which covers both generation and consumption.

> **"Municipalities report that suitable industrial plots are virtually sold out"**
> This single reported observation, if accurate, is the entire Atlas thesis for this candidate refuted in one line — and §13 provides the corridor-level confirmation. Atlas's site-origination model assumes a market where good sites are underpriced because nobody has yet recognised them. Finland in 2026 is the opposite market.

23. Competitive Pressure: Validation or Consumption?

The brief requires Atlas to determine whether existing projects **validate the market** or **consume the opportunity**. The answer is both, and the ratio is what matters.

Resource	Effect	Assessment
Power / grid queue	CONSUMED	Every committed project holds or seeks a connection position in a region under a stated connection restriction. Fingrid contracts against valid permits, so incumbents with permits sit ahead of Atlas, which has none. The queue is not first-come — it is first-permitted, which is worse for Atlas.
Land	CONSUMED	§13: four parcels confirmed taken, one contracted, one in preparation. The Koria plot moved from municipal ownership to Hyperco in 2024. Redeve monetised both of its mill estates.
District heating off-take	CONSUMED	§20: Hamina at c.80% of demand met by Google; KSS committed to atNorth and Hyperco. Only Kotka's network is unpartnered.
Contractors	CONSUMED	YIT holds contracts for both atNorth Myllykoski (c.€300m) and Hyperco Koria. Caverion holds the Kouvola MEP scope — described as one of the largest contracts in Assemblin Caverion Group history. Eltel holds the grid connection scope. The corridor's three most capable delivery partners are all contracted to Atlas's competitors through 2027. [FI02-MKT-004, -006; FI02-GRD-009]
Transformers	CONSUMED	Fingrid has itself contracted seven 400 MVA / 400 kV units from Hitachi Energy with an option for two more. [FI02-GRD-012] When the TSO is locking in multi-year transformer supply, a speculative developer is at the back of the queue.
Labour	CONSUMED	§31. atNorth reported roughly a hundred applications following a single Kouvola recruitment event, in a region losing population. Three simultaneous construction projects in one city of 77,625 people will bid up local wages.
Customer demand	VALIDATED	The one genuinely positive column. TikTok/ByteDance committed a €1bn tenancy in Kouvola. atNorth reports ongoing discussions with AI companies and hyperscalers. Demand for Finnish capacity is proven — **and it is being met by others' supply.**
Capital	VALIDATED	Partners Group, Damac/Edgnex, Brookfield plus a sovereign wealth investor, NREP, Varma, Alphabet. Institutional capital is available for Finnish data centres — for sponsors with track record.

Table 15. Resource-by-resource competitive assessment. Six of eight resources consumed; two validated. ***Validation without access is not an opportunity.***

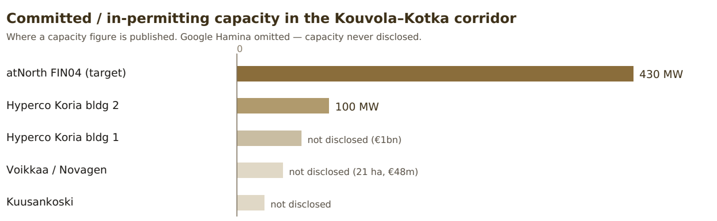


Figure 4. Published capacity alone exceeds 530 MW in a single municipality. Bars in pale tone indicate developments whose capacity is committed but undisclosed — the true total is higher.

24. European Benchmark

Atlas benchmarks the corridor against ten European locations on the dimensions that actually determine siting. Scores are Atlas judgements on a 1–5 scale from the evidence assembled in this dossier and Atlas's comparable work; they are not third-party ratings.

Location	Power price	Grid access	Carbon intensity	Free cooling	Land avail.	Latency to FRA	Note
Kouvola–Kotka	5	**1**	5	5	**1**	3	Excellent fundamentals; access foreclosed. The two 1s are the story.
Luleå / Boden, SE	5	3	5	5	4	2	The Nordic archetype. Farther from continental Europe; multiple bidding zones create locational price risk absent in Finland.
Stockholm, SE	3	2	4	4	2	4	Historic capacity constraints in the Stockholm area.
Oslo region, NO	4	3	5	4	3	3	Hydro-dominated; strong ESG profile; political scrutiny of power-intensive load rising.
Sines, PT	3	4	4	2	4	3	Atlantic cable landing plus deep-water port. The closest structural analogue to a coastal industrial conversion — and considerably less crowded.
Aragón, ES	3	4	4	2	5	3	Abundant land and renewables; hyperscaler cluster established. Cooling penalty in summer.
Madrid, ES	3	2	3	1	2	4	Major FLAP-adjacent hub; land and power both contested.
Frankfurt, DE	2	1	2	2	1	5	Latency benchmark; everything else adverse.
Paris, FR	4	3	5	2	2	4	Nuclear-backed low carbon; France pre-allocated 18 GW of grid capacity for data centres with a first-ready-first-served principle rolling out in 2026 and fast-tracks for strategic projects. [FI02-REG-011] **A materially more accessible grid regime than Finland's.**
Dublin, IE	2	1	2	3	1	4	Effectively closed to new large load in the Dublin region for years.
Warsaw / Poznań, PL	3	3	1	3	4	4	High carbon intensity is the binding constraint for ESG-committed operators.

Table 16. European benchmark. Atlas scores, 1–5, higher is better. Kouvola–Kotka scores at or near the top on every physical dimension and bottom on the two access dimensions.

The benchmark's most useful output

France's approach — pre-allocating 18 GW of grid capacity for data centres with published "powered sites" available for gigawatt-scale projects and a fast-track for strategic ones [FI02-REG-011] — is the regime an Atlas-scale developer can actually navigate, because it substitutes published state allocation for privately negotiated queue position. Atlas should weight **published capacity allocation regimes** heavily in future candidate screening. The absence of published node-level capacity data in Finland is not merely an information gap in this dossier; it is a structural barrier that systematically advantages incumbents with direct TSO relationships.

Regulation, Tax & Permitting

25. Permitting Roadmap

#	Approval	Authority	Timing	Note
1	Land control (purchase, lease or plot reservation)	Landowner / municipal council	Council decision cycle	Loviisa granted a plot reservation in autumn 2025 conferring the right to apply for permits and begin preconstruction; the lease and preliminary agreement followed on 21 January 2026. [FI02-LND-012]
2	MoD permit (non-EU/EEA acquirers >10%)	Ministry of Defence	Not established	Required for real estate purchase. [FI02-REG-010]
3	Zoning / detailed plan amendment (if needed)	Municipality	**Potentially years, with multiple appeal rounds**	Avoidable only by selecting a plot already zoned for the use. [FI02-REG-008] This is the strongest argument for brownfield industrial land and the strongest argument against the Savontalo listing.
4	Environmental impact assessment (EIA/YVA)	ELY Centre (regional)	Project-dependent	Applies to associated grid infrastructure; Fingrid runs full YVA procedures for 400 kV lines with ELY Centres as coordinating authority. [FI02-REG-012]
5	Environmental permit (water, discharge, noise)	Regional State Administrative Agency (AVI); appeals to Vaasa Administrative Court and ultimately the Supreme Administrative Court	**Can run to years**	Finnish environmental permitting can be protracted: one recent industrial permit process ran nearly ten years to a Supreme Administrative Court decision in February 2026. [FI02-REG-013] That case is not a data centre, and Atlas does not present it as typical — but it establishes the tail risk.
6	Building permit	Municipal technical committee	Weeks where zoning is correct	Hyperco's second Koria permit was filed in early June 2026 and scheduled for committee on 16 June 2026, with an early-start request. [FI02-LND-006]
7	Grid connection agreement	Fingrid	Gated on 3 and 6	Requires legally valid zoning and building permit. [FI02-GRD-019]
8	KJV2026 / large demand facility technical compliance	Fingrid; Energy Authority	In force for ≥30 MW direct connectees from 12 June 2026	See §8. [FI02-GRD-023]
9	Data centre registration	To be designated	Legislative project launched Dec 2025	New obligation; a **condition of state aid eligibility**. [FI02-REG-003]
10	Energisation and commissioning	Fingrid; DSO	—	—

Table 17. Permitting roadmap, site identification to operational data centre. Steps 1, 3 and 7 are the Atlas-blocking sequence.

26. Environmental Regulation

Area	Class	Assessment
EIA / YVA	CONFIRMED	Finland operates a structured YVA procedure with a coordinating ELY Centre issuing a reasoned conclusion. Fingrid transmission projects routinely run full YVA with route alternatives, Natura assessments, public consultation windows and published opinion registers. [FI02-REG-012] The process is transparent and predictable — but it takes time and it is public, which matters for confidentiality-sensitive tenants.
Water and discharge	UNKNOWN	Thermal discharge to the Gulf or Kymijoki would require permitting with water-body impact assessment. Not investigated at plot level.
Noise	UNKNOWN	Relevant for dry coolers and generator testing near residential areas. Voikkaa and Myllykoski are both adjacent to settlement.
Backup generation and fuel	UNKNOWN	Diesel storage and emissions permitting not investigated. Note that at 100–500 MW, backup generation is itself a significant permitted emission source.
Protected areas	UNVERIFIED	Natura 2000 designations feature prominently in Finnish grid EIAs. Not checked against corridor plots.
Contaminated land	UNKNOWN	See §15. Every large candidate parcel is a former pulp or paper mill. This is the largest unquantified environmental liability in the dossier.
Heritage	INDICATED	Sunila (Kotka) is an Alvar Aalto ensemble documented by DOCOMOMO Finland. [FI02-LND-015]

27. Taxation — Current to August 2026

This section is the most time-sensitive in the dossier and is stated with the three categories the brief requires kept strictly separate.

CURRENT LAW — in force at the date of this dossier

Provision	Rate	Detail
Electricity tax category I (general)	**2.24 c/kWh** energy tax	Data centres moved into category I **from 1 July 2026**. [FI02-REG-001, -002] CONFIRMED
Electricity tax category II (lower)	0.05 c/kWh energy tax	Remains available to qualifying industry — operations falling under main categories C or B of the Finnish Industry Classification, or subclass 38329 of category E. Determined by industry, not by consumption volume. Recipients must register as excise tax aid recipients. **Data centres no longer qualify.** [FI02-REG-002]
Strategic stockpile fee	0.085 c/kWh	Increased 1 April 2026. Applies to both categories. [FI02-REG-002]
Category I total incl. VAT (25.5%)	2.917875 c/kWh	Energy tax + stockpile fee + VAT. [FI02-REG-002]
Category II total incl. VAT	0.169425 c/kWh	For comparison. [FI02-REG-002]
Net change for data centres	**+2.19 c/kWh**	Government's own statement of the increase. Estimated to raise central government electricity tax revenue by **€47m annually at 2026 levels**; roughly half that in 2026 given the mid-year start. [FI02-REG-001]
Data centre state aid scheme	Capped at €30m annual revenue loss	Fixed-term tax relief for ten years based on electricity consumed. Legislation entering into force 1 July 2026. **Conditions:** the data centre must be registered under the new registration obligation, **and** must either reuse its waste heat or achieve a higher level of energy efficiency than required under the previous electricity tax relief legislation. [FI02-REG-003] CONFIRMED
Corporate income tax	Standard Finnish rate	Not the binding variable for this asset class.
Property tax	Municipal	Not established for corridor municipalities. UNKNOWN

PROPOSED CHANGE / UNRESOLVED AT THE DATE OF THIS DOSSIER

- **Detailed state aid terms.** At the December 2025 announcement the Government stated terms would be specified later, taking into account the need for compatibility with EU state aid rules. [FI02-REG-003] Atlas has **not** established whether final terms, eligibility thresholds or EU clearance were settled by August 2026. UNKNOWN — and this is a gate-blocking unknown for anyone underwriting today.
- **The aid gap.** Commentary in advance of the change noted a temporary gap in which operators face the higher tax without offsetting relief, with the replacement mechanism not expected to take effect until later in 2026. [FI02-REG-006] The Government's own December 2025 decision stated the legislation would enter into force on 1 July 2026, coincident with the tax rise. [FI02-REG-003] **These two statements are in tension — see contradiction C3 below.**
- **Registration obligation legislation.** A cross-administrative legislative project was launched in December 2025 to establish a data centre register and associated obligations, with responsibility for implementation to lie with an authority yet to be designated. [FI02-REG-003]
- **Mandatory matching generation.** A Kokoomus legislative proposal, reported as supported by the SDP, would require large data centres to obtain matching new power generation capacity before connecting to the Fingrid grid. [FI02-REG-007] INDICATED — status unverified.

> **Contradiction disclosed — C3: timing of the replacement aid scheme**
>
> Legal commentary from December 2025 described the new subsidy as "scheduled to enter into force in autumn 2026, immediately after the current electricity tax relief expires", likely taking a form other than a direct electricity tax reduction, with details still under preparation. [FI02-REG-005] The Prime Minister's Office press release of the same month states the legislation "will enter into force on 1 July 2026" and describes the aid as taking the form of fixed-term tax relief based on electricity consumed. [FI02-REG-003] **Atlas resolution:** the Government press release is Tier 1 and is preferred on both timing and form; the legal commentary was written before or contemporaneously with the Ministerial Committee decision and may reflect the earlier state of preparation. **However, Atlas cannot confirm the scheme was operational by August 2026**, and an operator underwriting today must treat relief as contingent, not assumed.

POLITICAL DISCUSSION

The reform was contested within the governing coalition. Finance Minister Riikka Purra announced the intention in March 2025 without prior consultation, prompting Finnish Energy's director to warn it would send a negative signal to energy-intensive industries. [FI02-REG-009] Fingrid's former CEO Jukka Ruusunen publicly welcomed the decision, telling Helsingin Sanomat there was no justification for the lower rate. [FI02-REG-009] The increase was described as a topic of heated debate among the governing parties, with the new support scheme forming part of the compromise. [FI02-REG-005] The SDP's deputy chair argued future data centres must contribute positively rather than drain finite energy resources, and that stricter obligations, greater flexibility and energy efficiency measures should be demanded of both greenfield and existing facilities. [FI02-REG-007]

Atlas reads the political direction as settled and one-directional. Government and opposition agree that data centres should pay more and contribute more; they differ on mechanism. A developer underwriting a twenty-year Finnish asset should assume the regulatory burden increases from here, not decreases.

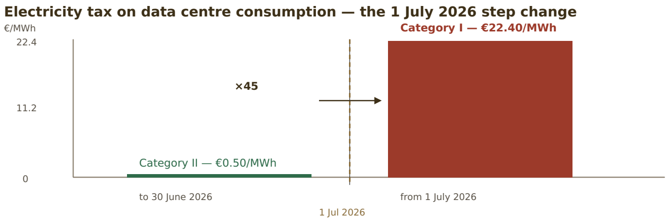


Figure 5. A 45-fold increase in the tax line, effective six weeks before this dossier's date. Excludes the strategic stockpile fee and VAT, and excludes any relief under the replacement aid scheme whose operational status Atlas could not confirm.

PART VIII

Delivery Capability

28. Construction

This is the corridor's second unambiguous strength — and, per §23, a strength currently working for Atlas's competitors.

Contractor	Role	Corridor evidence
YIT Corporation	Main contractor / design-build	Two live corridor data centre contracts. atNorth Myllykoski: comprehensive design-build covering design, CSA, MEP and commissioning, c.**€300m**, agreed July 2026, entering YIT's Q3 2026 order book, completion end-2027. Hyperco Koria: main contractor, ground broken end-October 2025, completion May 2027. Also contracted for XTX's second Kajaani building. [FI02-MKT-004, -006] CONFIRMED
Caverion (Assemblin Caverion Group)	Technical construction / MEP	Agreement signed 10 December 2025 with Hyperco Fin HoldCo 1 for technical construction and installation of data infrastructure at Kouvola — cooling, ventilation, fire alarm, fire extinguishing, smoke extraction and electrical systems. **Described as one of the largest contracts in Assemblin Caverion Group history**; value undisclosed by agreement. [FI02-MKT-006] CONFIRMED
Eltel	Grid connection EPC	c.**€16m** turnkey 110 kV substation plus cabling from the Hyperco site to the Fingrid substation. Works from summer 2025; Eltel scope planned complete 2026. Eltel employs c.4,500 across the Nordics, Germany and Lithuania. [FI02-GRD-009] CONFIRMED

The delivery ecosystem is proven in this exact corridor at this exact scale. Caverion's CEO for Finland framed the Kouvola award as evidence that Finnish expertise had convinced international investors and developers of large industrial and data infrastructure projects. Hyperco's co-founder cited the strengthening of domestic content and employment impact in the Kouvola region and across Finland. [FI02-MKT-006] An Atlas project would not face a capability gap. It would face a **capacity queue** behind three concurrent projects.

29. Supply Chain

Item	Risk	Assessment
Large power transformers	HIGH	The binding global constraint. Fingrid has contracted seven 400 MVA / 400 kV units from Hitachi Energy with an option for two more, manufactured at an expanding Hitachi Energy plant in the Vaasa region. [FI02-GRD-012] Domestic manufacturing capacity is a genuine Finnish advantage — but it is being absorbed by the TSO's own €5.2bn programme.
HV switchgear	MEDIUM	Not specifically investigated. Global lead times elevated.
Generators	UNKNOWN	Not investigated.
Chillers / dry coolers	MEDIUM	Caverion delivering cooling scope in-corridor, indicating supply chains are functioning.
Liquid cooling	MEDIUM	atNorth indicates DLC in new builds. [FI02-WTR-003] The specialist supply chain is thin globally.
Steel and concrete	LOW	Finland has established heavy construction supply chains; HaminaKotka handles dry bulk and project cargo at scale.
Modular / prefabricated systems	UNKNOWN	Not investigated.

30. Logistics

Logistics is a real corridor strength, well evidenced, and — unlike land and grid — not consumable by competitors.

Asset	Detail
Port of HaminaKotka	Finland's largest universal port, formed by the 2011 merger of the Kotka and Hamina ports, owned by HaminaKotka Satama Oy (jointly owned by the towns of Kotka and Hamina). Port area cited at c.1,100–1,200 ha. Finland's largest container port and terminal — Mussalo — with over ten container berths, roughly seven berths and seven gantry cranes at Mussalo, and annual container capacity cited at approximately one million TEU. Handles containers, RoRo, LoLo, dry bulk, liquid bulk, gas and **project shipments of all types**, and explicitly "specialises in handling of demanding project shipments". The port states its area is designed for long-term growth with abundant space for industrial investment and extension. Direct E18 motorway and rail access. Q1 2025 throughput 3,658,725 tonnes. [FI02-LOG-001, -002, -003] CONFIRMED
E18 motorway	Four-lane connection; Helsinki reachable in approximately 75 minutes from the Mussalo area; 6 km from the Mussalo site to the E18. [FI02-LND-009]
Rail	Kouvola is one of Finland's busiest railway crossing points with traffic in four directions, on the Riihimäki–Saint Petersburg line, c.130 km from Helsinki Central. Direct rail access to the port. [FI02-LOG-004]
Air	Helsinki Airport c.130 km / 1h15 by car from Mussalo. [FI02-LND-009]
Oversized equipment	The combination of a deep-water project-cargo port, direct motorway and rail, and a 52 km road distance from Hamina to Kouvola makes transformer and chiller delivery to inland sites straightforward. **This is a genuine and underrated corridor advantage.**

31. Workforce

Atlas treats workforce as a **risk** in this corridor, against the regional marketing position that "there is plenty of skilled labour available".

Factor	Assessment
Regional population trend	ADVERSE Kymenlaakso fell from 186,030 (1990) to 156,198 (2025) — 16% over 35 years, 4% in the last five. [FI02-WRK-001] The Xamk figure of "a population of 185,000" used in student-facing material is materially out of date. **Contradiction C4, resolved in favour of Statistics Finland.**
Higher education	POSITIVE South-Eastern Finland University of Applied Sciences (Xamk) is Finland's third largest UAS, c.12,000–12,300 students and over 1,000 staff across Mikkeli, Kotka, Kouvola and Savonlinna, jointly owned by those four cities. The Kotka campus has c.200 staff and c.3,500 students. Xamk offers a Master of Engineering in **Digital Industry and AI** and a Master's in Strategic Supply Chain Management. Roughly 2,400 graduates in 2024. [FI02-WRK-008, -009] CONFIRMED No research university in the region.
Demonstrated recruitment	POSITIVE atNorth's Finland operational lead reported roughly **a hundred applications** following a single Kouvola recruitment event in autumn 2025. Phase one of FIN04 is stated to employ around sixty people, with atNorth's centres later employing around two hundred. [FI02-WRK-007] CONFIRMED
Construction workforce	CONTESTED Google's Hamina expansion alone was projected by a Deloitte impact study to support approximately 1,900 direct, 1,650 indirect and 640 induced construction jobs annually over two years. [FI02-WRK-010] DayOne's Lahti campus projects 1,000 construction workers at peak. [FI02-MKT-009] Against a regional population of 156,198 and three concurrent Kouvola projects, this is a genuine labour market squeeze.
Operating headcount	Google Hamina employs around 400–500 in roles spanning computer technicians, engineers, security, catering and facilities management. [FI02-WRK-011] Data centres are capital-intensive and labour-light; the regional development case rests mainly on construction and tax base.
Salaries and competition	UNKNOWN No salary data obtained. With three concurrent projects competing for the same technicians in a shrinking labour pool, upward pressure should be assumed.

Table 18. Workforce assessment. The education infrastructure is genuinely good; the demographic base is genuinely shrinking; and the corridor is about to run three large projects through it simultaneously.

PART IX

Commercial Demand

32. Who Could Actually Use the Site?

NO OPERATOR DEMAND VALIDATED

Nothing in this section constitutes evidence of interest in an Atlas-originated site. Every entry describes an operator's general market behaviour, not a signal directed at Atlas. Atlas sits at **Level 0** on the ladder in §34.

Operator	Finland presence	Relevance	Evidence and Atlas assessment
ByteDance / TikTok	**Committed — Kouvola**	Already served	€1bn investment in a Kouvola data centre built by HyperCo, described as a core part of Project Clover, TikTok's €12bn European data protection initiative for its 175 million European users. Cited a strong digital infrastructure, clean energy mix, robust data governance and skilled tech talent as making Finland and Kouvola an ideal location. [FI02-DEM-001] CONFIRMED **The most important demand signal in the corridor — and it is fully allocated.**
Google	**Operating — Hamina**	Self-supplied	Fifteen years of operation, seven facilities, own land bank. Builds and owns; does not take third-party colocation at this scale. Has publicly linked future Finnish expansion to the tax settlement. [FI02-REG-006] **Not an Atlas customer under any scenario.**
Microsoft	Present in Finland	Theoretical	Cited among operators with a Finnish presence; acquired 19 ha on Finland's west coast for a data centre. [FI02-MKT-018] INDICATED No corridor activity identified.
Nebius	Lappeenranta (with Polarnode)	Direct AI competitor, adjacent	AI-focused campus at Pajarila, Lappeenranta, c.87 km east of Kouvola. [FI02-MKT-012] INDICATED Nebius has chosen its South-East Finland location and it is not this corridor.
CoreWeave, Nscale, Crusoe, Fluidstack	None identified in Finland	Theoretical	**Atlas found no evidence of Finnish activity by any of these operators.** They are named in the brief as plausible AI infrastructure customers; Atlas will not manufacture relevance where the evidence is absent. UNKNOWN
AWS, Meta, Oracle	None identified in corridor	Theoretical	No corridor evidence found. UNKNOWN
DataCrunch	Three Helsinki facilities; proposed Akaa	Instructive	Finnish-founded AI cloud company; deployed an H200 GPU cluster in Finland; raised $13m seed in October 2024. Its Akaa proposal was reportedly affected by Fingrid's connection pause for projects needing 10 MW or more. [FI02-DEM-002] **A directly comparable small AI operator hitting exactly the constraint identified in §7.**
Sovereign AI programmes	Finland is an EuroHPC host nation	Theoretical	Finland's national supercomputing centre CSC hosts EuroHPC infrastructure at Kajaani, not in this corridor. Sovereign AI capacity in Finland has an established northern locus. UNVERIFIED
atNorth's own pipeline	Kouvola	Competitor, not customer	atNorth's CEO has stated the company is in "ongoing discussions with several AI companies, hyperscalers as well as international organizations requiring high-performance computing". [FI02-MKT-002] This is real AI demand for the corridor — **routed to a competitor's 430 MW campus.**

Table 19. Potential users. Two operators have committed to the corridor; both are served by others. Six are theoretical with no Finnish or no corridor evidence.

33. Operator Requirement Matrix

Atlas reverse-engineers operator requirements from published behaviour. The rightmost column is the discipline: **PUBLIC** means the operator stated or demonstrated it; **INFERENCE** means Atlas deduced it and it may be wrong.

Operator	Initial MW	Expansion	Land	Power	Cooling	Fibre	Geography	Basis
ByteDance / TikTok	Not disclosed	Multi-phase	24 ha via developer	Clean energy mix stated as a criterion	Waste heat integration stated	Not stated	EU data residency required	**PUBLIC** — TikTok newsroom [FI02-DEM-001]
Google	—	Seven facilities to date	Owns; 50 ha bank	Contracted CFE; wind PPAs; 24/7 CFE by 2030 target	Seawater; PUE c.1.1; offsite heat recovery	Own network	Coastal, seawater access	**PUBLIC** [FI02-WTR-001, -002; FI02-PWR-016]
DayOne	50 MW first building	128 MW site; 281 MW Finland	98,901 m² fully zoned industrial	Renewable grid	Zero freshwater; waste heat to Lahti Energy	Not stated	"Crossroads of Nordic and continental connectivity"	**PUBLIC** [FI02-MKT-009, -010]
atNorth	60 MW phase 1	430 MW target	21 ha → 45 ha	Renewable	Free air cooling; DLC in new builds; heat reuse via KSS	Not stated	Nordic; low population density; land open for development	**PUBLIC** + *INFERENCE* on DLC [FI02-MKT-002, -004; FI02-WTR-003]
Hyperco / Edgnex	Not disclosed	100 MW second building; 300 MW at Lohja	24 ha; 130 ha at Pyhäjoki	110 kV customer substation to Fingrid	Caverion MEP scope	Not stated	Multi-site Finland	**PUBLIC** [FI02-MKT-003, -006, -007]
Generic AI training	*50–100 MW*	*To 250–500 MW*	*20–45 ha*	*Dual feed; ride-through*	*DLC at 100 kW+/rack*	*Two diverse routes, sub-*	*Low-carbon, stable*	***ATLAS INFERENCE** — not observed*

operator				*compliant*		*25 ms to FRA*	*jurisdiction*		

Table 20. Operator requirement matrix. Note how little is publicly stated about fibre requirements — a gap Atlas cannot close by desk research.

34. Demand Validation Ladder

Atlas defines demand validation to prevent the single most common failure mode in infrastructure development: mistaking a conversation for a customer.

Level	Definition	Atlas status	What it would take
Level 0	**No contact**	**← ATLAS IS HERE**	—
Level 1	Operator conversation held	Not achieved	A meeting with a named individual with siting authority. Requires a credible site to discuss.
Level 2	NDA executed; technical data exchanged	Not achieved	Requires a site data pack: grid position, land control, fibre, water, permitting status.
Level 3	Written expression of interest	Not achieved	Non-binding but written. **The Atlas Gate 2 threshold.**
Level 4	Letter of intent	Not achieved	Commercially meaningful; typically requires site control.
Level 5	Capacity reservation or development agreement	Not achieved	Bankable.

Atlas must not describe a conversation as customer demand. Nor may it describe TikTok's Kouvola commitment, atNorth's discussions with AI companies, or DayOne's tenant negotiations as evidence of demand for an Atlas site. Those are competitor achievements. Restating them as market validation in an Atlas investor document would be materially misleading, and this dossier records the prohibition explicitly so that it cannot be softened later.

Stakeholders & Outreach

35. Stakeholder Map

Category	Organisation	Role and relevance
Transmission	Fingrid Oyj	TSO. Holds the answer to the only question that matters: node-level capacity at Koria and Kymi. Runs the connection agreement process, the KJV2026 grid code, and the Grid Scope capacity map service.
Regulator	Energy Authority (Energiavirasto)	Confirms Fingrid grid code specifications; economic regulation of networks.
Distribution	Kymenlaakson Sähköverkko; municipal network companies	Regional distribution below the Fingrid boundary. Relevant for sub-100 MW connections.
Municipalities	City of Kouvola; City of Kotka; City of Hamina; City of Loviisa (adjacent)	Landowners, zoning authorities, building permit authorities, and shareholders in the local energy and port companies. Kouvola sold the Koria plot to Hyperco in 2024.
Regional	Regional Council of Kymenlaakso; South-East Finland ELY Centre	Regional land use planning; EIA coordinating authority.
National promotion	Business Finland; Invest in Finland	Operates the national data centre site catalogue and map used in §13. Publicly celebrated the Google Hamina expansion.
Regional development	**Cursor Oy** (Kotka–Hamina, operating the "Power Coast" cluster); **Kouvola Innovation Oy / Kinno**	The two regional development agencies. Cursor worked with the City of Loviisa for several years to attract green transition investment and brokered the Nova Complex arrangement. These are the correct first-contact points for any corridor engagement.
Energy utilities	KSS Energia (Kouvola); Kotkan Energia; Haminan Energia; Lahti Energy (adjacent)	District heating off-take counterparties. Two of three corridor networks are already partnered.
Landowners	**Redeve**; City of Kouvola; City of Kotka; atNorth; Hyperco; Alphabet/Tuike Finland Oy	Redeve is the pivotal private landowner: it owned both the Myllykoski and Voikkaa mill estates, sold 21 ha to atNorth, agreed c.21 ha to Novagen at €48m, and states it retains more than 50 ha at Myllykoski.
Port	HaminaKotka Satama Oy	Jointly owned by Kotka and Hamina. Project cargo and heavy equipment logistics.
Fibre	Cinia Oy (majority state-owned); Telia Finland; Elisa; DNA; GlobalConnect; Lounea	Route diversity and dark fibre availability — the largest unresolved technical question after power.
Construction	YIT; Caverion; Eltel	All three currently contracted to corridor competitors.
Incumbent operators	Google; atNorth; Hyperco/Edgnex; DayOne; Novagen; Nova Complex	Competitors — and, in a non-origination model, potential counterparties.
Investors	Partners Group; Damac/Edgnex; Brookfield; NREP; Varma	The capital already deployed. Atlas's realistic ceiling is to be a service provider to this group, not a peer.
Government	Ministry of Finance; Ministry of Economic Affairs and Employment; Prime Minister's Office; Ministry of Defence	Tax, state aid scheme, registration obligation, real estate acquisition permits.
Industry bodies	Finnish Data Center Association (FDCA); Finnish Energy; Confederation of Finnish Industries	Market data and policy advocacy.
Education	Xamk (South-Eastern Finland UAS)	Workforce pipeline; Digital Industry and AI master's programme.

36. Named Contact Research

Method note — and a deliberate departure from the brief

The brief requests 30–50 named individuals with contact details. Atlas has **not** produced that list, for two reasons stated plainly.

First, it would be dishonest. Atlas cannot verify a single personal email address from desk research. Producing a table of thirty inferred addresses formatted to look like a contact database — even with an "INFERRED" flag — creates an artefact that will be used as though it were verified the moment it leaves this document. The brief itself forbids representing inferred emails as verified; the safest way to honour that is not to generate them.

Second, it would be pointless. Gate 1 is a FAIL. A fifty-person outreach list for a candidate Atlas is not pursuing is wasted work and, if sent, wasted goodwill with counterparties Atlas may want later. What follows instead is a short list of **named individuals already on the public record in this corridor**, given in their professional capacity, with organisational contact routes only — plus the two organisations that are the correct doors to knock on.

Name	Organisation	Position	Relevance	Contact route
Marita Toikka	City of Kouvola	Mayor / City Manager	Publicly welcomed both the atNorth and TikTok/HyperCo investments; the city sold the Koria plot.	Municipal switchboard. **Personal email: NOT VERIFIED — do not infer.**
Marko Riipinen	KSS Energia	CEO	The corridor's key district heating counterparty; publicly committed to the atNorth heat reuse	Company published channels. **NOT VERIFIED**

			arrangement.	
David Lindström	Cursor Oy	CEO	Regional development agency for Kotka–Hamina; brokered the Nova Complex/Loviisa arrangement.	**Cursor Oy organisational contact — the correct first door. NOT VERIFIED**
Jari Lahtinen	Redeve	Managing Director	Signed the atNorth Myllykoski land agreement. Redeve retains >50 ha — the only plausibly available large parcel.	Company published channels. **NOT VERIFIED**
Jussi Jyrinsalo	Fingrid Oyj	Senior Vice President, Grid Services and Planning	Named in Fingrid media releases on connection capacity and the new connection information procedure. **The right function for the capacity question.**	Fingrid media/customer channels. **NOT VERIFIED**
Petri Parviainen	Fingrid Oyj	Unit Manager, Main Grid Services	Publicly stated the southern Finland connection position.	Fingrid customer channels. **NOT VERIFIED**
Taina Ahti-Aalto	atNorth	Finland operational lead	Public on FIN04 build progress and Kouvola recruitment.	Company channels. **NOT VERIFIED**
Aleksi Taipale	Hyperco	Co-founder and CEO	Signed the Caverion agreement; leads the corridor's largest tenanted project.	Company channels. **NOT VERIFIED**
Kimmo Vaarala	Hyperco	Project Manager, Koria	Public on Koria construction status.	Company channels. **NOT VERIFIED**
Ville Tamminen	Caverion Finland	CEO / EVP	Delivering the Kouvola MEP scope.	Company channels. **NOT VERIFIED**
Jamie Khoo	DayOne Data Centers	CEO	Adjacent Lahti campus; Kouvola partnership with Hyperco.	Company channels. **NOT VERIFIED**
Bernard Mah	Nova Complex	COO	Loviisa project; a comparable smaller international entrant.	Company channels. **NOT VERIFIED**
Veli-Matti Mattila	Appointed by the Prime Minister	Rapporteur, National Roadmap for Data Centres	Authored the roadmap on which the state aid scheme is based.	Via Prime Minister's Office. **NOT VERIFIED**
Mikko Martikkala	Prime Minister's Office	Special Adviser to the PM in Economic Affairs	**Named as the inquiries contact on the state aid decision release.**	Published in the Government press release; the only contact in this table with a publicly published direct line. Use the official published channel.
Ari-Jussi Knaapila	Cinia Oy	CEO	C-Lion1 and the national fibre backbone.	Company channels. **NOT VERIFIED**

Table 21. Named individuals on the public record. Fifteen names, all in professional capacity, all sourced from press releases, government publications or trade coverage. ***Zero email addresses asserted.*** *Atlas will verify contact details at the point of genuine engagement, not before.*

37. Contact Strategy

Atlas would sequence outreach in four waves. **None of these waves should be executed for this candidate.** The sequence is documented because it is the reusable methodology, and because the reasoning explains why Gate 1 stops here.

Wave	Purpose	Counterparties	Why this sequence
Wave 1 Infrastructure truth	Establish whether the physical opportunity exists at all	Fingrid (capacity); Cursor Oy and Kouvola Innovation (land and municipal appetite); Redeve (retained estate)	**Always first, and never skippable.** Every subsequent wave costs credibility if the infrastructure answer turns out to be no. A developer who approaches an operator before knowing its grid position is asking to be dismissed permanently.
Wave 2 Technical feasibility	Convert infrastructure truth into a costed, engineered proposition	Cinia, Telia (fibre and route diversity); municipal water undertakings; KSS Energia and Kotkan Energia (heat); YIT, Caverion, Eltel (delivery)	Only meaningful once a specific plot and connection point exist. Approaching carriers without a plot yields generic answers with no commercial value.
Wave 3 Demand	Move from Level 0 toward Level 3	AI infrastructure operators and hyperscalers	**Third, not first.** The instinct to lead with demand is the classic error: an operator asked to express interest in a site with no grid position and no land control will decline and will remember. Atlas gets one credible approach to each operator.
Wave 4 Capital	Fund development	Infrastructure funds; strategic partners	Last. Capital prices the package assembled in waves 1–3. Approaching capital first produces a valuation of nothing.

Why the sequence itself proves the Gate 1 decision. Atlas has run Wave 1 to the limit of desk research and hit a wall on all three counterparties: Fingrid's answer is unobtainable publicly and Atlas has no standing to ask formally; the land is taken; the one retained estate belongs to a party transacting at €48m with institutional buyers. Wave 1 has failed at the desk stage. Waves 2, 3 and 4 are unreachable, and executing any of them out of sequence would burn relationships Atlas may need in a different geography.

Risks & Validation

38. Full Risk Register

ID	Risk	Prob.	Impact	Evidence	Mitigation	Owner	Gate effect
R01	**No available grid connection capacity at corridor nodes**	High	Fatal	Fingrid southern Finland restriction 2025–2027; no published node capacity [GRD-006,-007]	Written Fingrid confirmation — Atlas has no standing to obtain it	Fingrid	**GATE-BLOCKING**
R02	**Land unavailable / unaffordable to Atlas**	Very high	Fatal	§13 — four parcels taken, one contracted at €48m [LND-001..012]	None identified	Atlas	**GATE-BLOCKING**
R03	**Electricity tax reclassification degrades opex**	Certain	High	In force 1 July 2026; +2.19 c/kWh [REG-001]	State aid scheme — capped, conditional, terms unconfirmed	MoF / TEM	**GATE-BLOCKING**
R04	**No operator demand for an Atlas site**	Certain today	Fatal	Level 0, §34	Wave 3 outreach — unreachable without R01/R02 resolved	Atlas	**GATE-BLOCKING**
R05	**Competitive saturation consumes remaining headroom**	Very high	Fatal	Six developments, §21, §23	None	Market	**GATE-BLOCKING**
R06	**State aid terms exclude Atlas-type projects**	Medium	High	Terms unspecified; conditional on registration + heat reuse/efficiency [REG-003]	Monitor; design for heat reuse from inception	TEM	**GATE-BLOCKING**
R07	Zoning amendment required and protracted	Medium	High	"Often lengthy, potentially years, multiple appeals" [REG-008]	Select pre-zoned industrial land only	Municipality	Major
R08	Brownfield contamination liability	Medium-high	High	All large parcels are former mills; no ESA seen	Phase I/II ESA before any option exercise	Atlas	Major
R09	Baltic subsea cable disruption	Medium	Medium	C-Lion1 severed 18 Nov 2024; characterised as sabotage by German defence minister [FBR-008]	Multi-route contracts; national-level diversity exists	Carriers	Moderate
R10	Storage/BESS mitigation unavailable	Certain to 2029	Medium	No new >5 MW storage in southern Finland before 2029 [GRD-013]	None available in-region	Fingrid	Moderate
R11	KJV2026 compliance cost unquantified	Certain	Medium	Fault ride-through and intake limitation for ≥30 MW from 12 June 2026 [GRD-023,-024]	Early electrical engineering input; monitor Energy Authority confirmation	Atlas / Fingrid	Moderate
R12	Mandatory matching generation legislation enacted	Medium	High	Kokoomus proposal, SDP-supported [REG-007]	Model a generation-plus-load structure	Parliament	Major
R13	No proven diverse hyperscale fibre to a specific plot	Medium	High	§16–17; C-Lion1 lands at Helsinki/Hanko not in-corridor	Carrier engagement in Wave 2	Carriers	Major
R14	Water abstraction/discharge permit refused or constrained	Low-medium	Medium	§19 all UNKNOWN	Zero-freshwater / air-side design	AVI	Moderate
R15	District heating off-take unavailable (network already served)	High	Medium	Hamina c.80% served by Google; KSS committed [HEA-001,-002]	Kotkan Energia is the only unpartnered network	Utilities	Major — and now a state aid condition
R16	Transformer and long-lead equipment shortage	High	High	Fingrid itself contracting 7+2 × 400 MVA units [GRD-012]	Early reservation; requires capital Atlas lacks	Supply chain	Major
R17	Contractor capacity queue	High	Medium	YIT, Caverion, Eltel all contracted to competitors through 2027	Engage second-tier or international EPC	Market	Moderate
R18	Labour scarcity and wage inflation	Medium-high	Medium	Population −16% since 1990; three concurrent projects [WRK-001]	Xamk partnership; import specialist labour	Atlas	Moderate
R19	Security/foreign ownership scrutiny	Medium	Medium	MoD permit for non-EU/EEA >10%; security authorities reviewing DC sector [REG-003,-010]	EU-domiciled structure; early engagement	MoD	Moderate
R20	Heritage constraint (Sunila)	Low	Medium	Alvar Aalto ensemble, DOCOMOMO-documented [LND-015]	Avoid Sunila	Heritage authority	Minor
R21	Environmental permitting tail risk	Low-medium	High	A recent Finnish industrial permit ran c.10 years to SAC decision [REG-013]	Avoid water-intensive designs; pre-consult ELY/AVI	AVI / courts	Major

R22	Regional demographic decline undermines the socio-economic case	Certain	Low	156,198 in 2025, falling [WRK-001]	None; note it cuts both ways — decline is why municipalities are welcoming	—	Minor
R23	Market cools; AI capex cycle turns	Medium	High	Forecasts range 1.5 GW to 5 GW by 2030–31 — a spread that signals genuine uncertainty [MKT-019..022]	Phased build; anchor tenant before construction	Market	Major
R24	Atlas methodology is correct but non-proprietary	Certain	Strategic	§12 — the just-transition thesis executed here by four separate parties	Apply earlier in the cycle in less-covered geographies	**Atlas**	**Programme-level**

Table 22. Risk register — 24 material risks, six of them gate-blocking. R24 is a finding about Atlas itself rather than about this corridor, and is the most valuable output of the exercise.

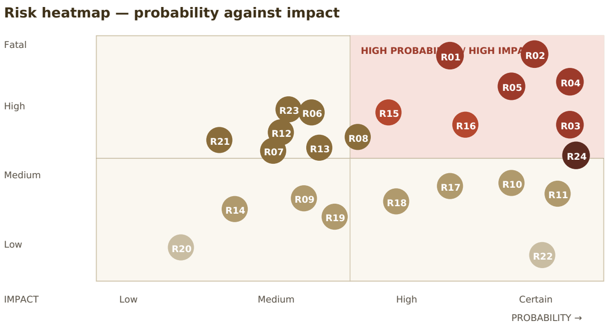


Figure 6. Risk heatmap. Five risks sit in the fatal/high-probability corner, which is not a research programme — it is a decision.

39. Red Team: Why Kouvola–Kotka May Be a Bad Data Centre Location

This section is written adversarially. Its purpose is to destroy the investment thesis. It succeeds.

1. It is not a location problem — it is an access problem, and access problems do not yield to effort

The most seductive error available to Atlas here is to read the corridor's genuine quality as an argument for persistence. It is not. Every quality Atlas can identify — clean power, cold climate, brownfield land, heat networks, a port, a skilled contractor base — is **visible to everyone** and has been acted on by parties with more capital, better relationships and a four-year head start. There is no informational edge to be had. Atlas would be entering a fully-priced market as its least-credentialed participant.

2. The grid is closed and the closure has already been extended once

Fingrid restricted new large demand connections in southern Finland through 2027. When reinforcement for storage slipped, Fingrid extended that restriction to 2029. [GRD-013] The base case for a developer entering now is not "the restriction lifts in 2027"; it is "the restriction lifts when Fingrid says it does, and Fingrid has already demonstrated willingness to extend." Meanwhile, Fingrid contracts capacity against valid permits, so the queue rewards those who already hold land — precisely the parties Atlas is behind.

3. Everything worth having in the corridor is already taken

Six developments. Four confirmed-taken land parcels. Two of three district heating networks partnered. All three principal contractors committed. Restating §23: six of eight critical resources are consumed. The two that remain — customer demand and capital — are available to *sponsors with track record*, which is a category Atlas is not in.

4. The economics changed on 1 July 2026, mid-cycle

A 45-fold increase in the electricity tax line, worth roughly €19m a year on a 100 MW campus, landed six weeks before this dossier. The offsetting scheme is capped at €30m nationally — small relative to a sector heading toward 1.5 GW — is conditional on registration plus heat reuse or above-baseline efficiency, and its detailed terms and EU state aid compatibility were unresolved when announced. Incumbents with signed EPC contracts and anchor tenants absorb this. A speculative developer underwriting today cannot price it.

5. There is no anchor customer, and the plausible ones are spoken for

TikTok chose HyperCo. atNorth is talking to AI companies about its own 430 MW. DayOne is negotiating tenants for Lahti. Google builds for itself. Nebius went to Lappeenranta. Atlas has no relationship with any operator and nothing to offer one that a competitor cannot offer better and sooner.

6. Labour is thin and about to get thinner

Kymenlaakso has lost 16% of its population since 1990 and 4% in the last five years. Into that shrinking base the corridor is about to run three concurrent large construction projects, on top of a Google expansion projected to support around 1,900 direct construction jobs annually. Wage inflation and schedule risk should be assumed, not hoped against.

7. The connectivity claim is weaker than the marketing suggests

C-Lion1 does not land in this corridor. Traffic to Rostock traverses terrestrial fibre to Helsinki or Hanko first. Plot-level diverse hyperscale routing is unproven. And the Baltic has a demonstrated cable-severance risk that a German defence minister publicly called sabotage.

8. Geopolitics cuts both ways

Finland's NATO membership and rule-of-law stability are genuine assets. But this corridor sits on the eastern edge of the EU: Kotka is roughly 270 km from the St Petersburg region, the Russian border closure has materially damaged the regional economy in logistics, forestry and tourism [MKT-014], and Google's Hamina site has been described in international coverage as close to the Russian border. Some operators will read the location as strategically robust; others will read it as exposed. Atlas cannot assume the favourable reading.

9. Environmental liability is unquantified and structurally likely

Every large candidate parcel is a former pulp or paper mill. Atlas has seen no environmental site assessment for any of them. Historic mill contamination is common and remediation can be a multi-million-euro liability. A developer buying at €48m for 21 ha who has not done this work is taking a risk Atlas would not take — and a developer who has done it has information Atlas does not.

10. The regulatory direction is one-way

Government and opposition agree data centres should pay more and contribute more. A pending proposal would require large data centres to secure matching new generation before connecting. A registration obligation is being legislated. Security authorities are reviewing the sector. None of this is disqualifying; all of it raises the entry barrier in favour of incumbents.

Red team verdict

The thesis does not survive. The strongest available counter-argument in the corridor's favour is that the Redeve retained estate at Myllykoski exceeds 50 ha and that Kotkan Energia's district heating network has no data centre partner. Those two facts are the residue of an otherwise total refutation, and neither of them is worth anything without grid capacity Atlas cannot obtain and land Atlas cannot afford. **Atlas should not attempt to rescue this candidate.**

40. Critical Unknowns

#	Question	Why it matters	Who can answer	Evidence required	Impact if negative
1	**What uncommitted demand-connection capacity exists at Koria and Kymi?**	The single fact on which everything else depends	Fingrid	Written statement or Grid Scope extract at node level	**Candidate dead.** Already treated as the base case.
2	Does the 2025–2027 restriction bind at corridor nodes, and will it extend?	Determines whether any timeline exists	Fingrid	Written confirmation	Candidate dead
3	Is any parcel >20 ha optionable by Atlas at a price Atlas could pay?	No land, no project	Redeve; City of Kotka; City of Kouvola	Written option or plot reservation	Candidate dead. Base case.
4	What are the final state aid terms and did they receive EU clearance?	Determines whether a 100 MW campus pays €19m/yr or materially less	MoF; TEM; DG COMP	Published scheme rules	Opex model unfinanceable
5	Will matching-generation legislation be enacted?	Converts a DC developer into a generation developer	Eduskunta	Bill status	Business model change
6	Would any operator engage with an Atlas-originated corridor site?	Level 0 → Level 1	Operators	A meeting	No customer, no project
7	Does Redeve intend to sell the retained >50 ha, and to whom?	The only plausibly available large parcel	Redeve	Direct enquiry	Last land option closes
8	Is contamination present at Voikkaa, Myllykoski or Sunila, and at what remediation cost?	Could exceed land value	ELY Centre; Phase II ESA	Environmental site assessment	Parcel uneconomic
9	Do two physically diverse fibre routes exist to any corridor plot, at what cost and lead time?	Hyperscale tenancy precondition	Cinia; Telia; GlobalConnect	Carrier route maps and quotes	Ineligible for hyperscale tenancy
10	What is Kotkan Energia's heat off-take appetite and network absorption capacity?	The only unpartnered network; now a state aid condition	Kotkan Energia	Technical and commercial term sheet	Loses the aid qualification route
11	What is the current status of the Kotka Mussalo 26 ha campus reservation?	Last identified municipal parcel	City of Kotka	Direct enquiry	No coastal option
12	What water abstraction and thermal discharge would be permitted, and where?	Determines cooling architecture	AVI; ELY; municipal water undertakings	Pre-application consultation	Forces air-side design, higher opex
13	What does KJV2026 compliance cost at 100–500 MW?	New capex line since June 2026	Fingrid; electrical consultants	Engineering study	Capex model stale
14	What is the actual Fingrid transmission tariff for a direct connectee at this scale?	One of three UNKNOWN lines in the cost stack	Fingrid	Published tariff extract	Delivered cost understated
15	What transformer lead times apply to a non-TSO buyer?	Governs the critical path	Hitachi Energy; Siemens Energy; GE Vernova	Indicative quotations	Schedule slips years
16	Is EPC capacity available after 2027?	YIT, Caverion, Eltel all committed	Contractors	Capacity discussion	Schedule and cost risk
17	What property tax rate applies in Kouvola and Kotka to this asset class?	Unmodelled opex line	Municipalities	Published rates	Minor
18	Do historic mill water abstraction rights transfer with land title?	Could be a decisive hidden asset — it was for Google at Hamina	Finnish legal counsel	Title and permit review	Loses a potential edge
19	What is the corridor's actual electricity consumption and how much industrial load was decommissioned?	Would indicate whether legacy capacity was retained or reallocated	Fingrid; Statistics Finland; DSO	Regional consumption data	Removes the "exceptional legacy infrastructure" claim
20	Would Business Finland or Cursor Oy actively support an Atlas entry, or steer to incumbents?	Determines whether Atlas gets a hearing at all	Business Finland; Cursor Oy; Kouvola Innovation	Introductory meeting	No local sponsorship

Table 23. Twenty critical unknowns, ranked. The first six are gate-blocking; five of those six already have a working negative answer.

41. 30-Day Validation Programme

This programme should NOT be executed

It is specified because the brief requires it and because the template is reusable. Atlas's recommendation is to reallocate this budget to a candidate where questions 1 and 3 of §40 have plausible affirmative answers. **If the Investment Committee overrides the Gate 1 FAIL, days 1–5 alone are sufficient to confirm or reverse the decision, and no work beyond day 5 should be authorised until they return.**

Day	Counterparty	Contact route	Question	Evidence requested	Success criterion	Gate impact
1–3	**Fingrid** — Grid Services and Planning	Customer connection enquiry channel	What uncommitted demand connection capacity exists at Koria and Kymi, and does the 2025–2027 southern Finland restriction bind there?	Written statement or Grid Scope node extract	Any written figure, even zero	**DECISIVE**
3–5	**Redeve**	Company channels	Is the retained >50 ha at Myllykoski available, on what terms, and is it under exclusivity?	Indicative terms or a clear no	Written indication of availability	**DECISIVE**
5	**DECISION POINT — if either of the above returns negative, stop. Do not proceed to day 6.**					
6–8	City of Kotka	Business Kotka-Hamina / Cursor Oy	Current status of the Mussalo 26 ha data centre campus plot	Plot status and reservation position	Plot confirmed available	High
8–10	Cursor Oy; Kouvola Innovation	Direct organisational contact	Would the region support a new entrant, and what parcels are genuinely uncommitted?	Site list with current status	Named uncommitted parcel	High
10–13	Kotkan Energia	Company channels	Heat off-take appetite, network absorption capacity, temperature requirements, seasonality	Technical parameters	Stated MW-thermal absorption capacity	Medium — state aid relevant
13–16	Cinia; Telia	Wholesale/carrier channels	Do two physically diverse routes exist to named corridor locations? Lead time and indicative cost?	Route diversity confirmation	Two diverse routes confirmed	Medium
16–19	Ministry of Finance / TEM	Published channels; PMO adviser named in the state aid release	Final state aid scheme terms, eligibility, and EU state aid status	Scheme rules or timetable	Published terms obtained	High
19–22	ELY Centre South-East Finland	Pre-application consultation	Contamination register status for Voikkaa, Myllykoski, Sunila; water and discharge expectations	Register extracts; informal view	No disqualifying contamination	Medium
22–25	Finnish legal counsel	Engagement	Do historic mill water abstraction rights transfer with title? MoD permit implications for the capital structure	Written opinion	Clear answer	Medium
25–27	Electrical consultant	Engagement	Indicative KJV2026 compliance cost at 100 MW and 250 MW	Order-of-magnitude estimate	Costed range	Medium
27–30	Atlas internal	—	Consolidate; re-score; Gate 1 re-decision	Updated evidence register	Decision confirmed or reversed	—

Table 24. Thirty-day validation programme with a hard stop at day 5. Note that the programme contains no operator outreach: Wave 3 is unreachable until days 1–5 return positive.

Investment Conclusion

42. Bull Case

The bull case requires a specific and improbable conjunction of events. Atlas states it fully so that it can be tested rather than dismissed.

The chain. Fingrid confirms in writing that reinforcement completing in 2027 releases material demand-connection headroom at Koria, and that the corridor's committed projects have not absorbed it. Redeve confirms its retained Myllykoski estate — more than 50 hectares — is available, and prices it at a level a pre-revenue developer can option rather than buy. Kotkan Energia, the corridor's one unpartnered district heating utility, offers heat off-take terms that qualify a project under the new state aid scheme, restoring a meaningful share of the 2.19 c/kWh tax delta. The state aid scheme's final terms are published, are generous to efficient heat-reusing projects, and clear EU state aid review. Cinia confirms two physically diverse fibre routes at commercially viable cost. And an AI infrastructure operator — most plausibly one currently without Finnish capacity — engages, because the corridor's remaining land is the last available parcel adjacent to a 400 kV anchor in a market where everything else is gone.

What it would be worth. A 100–250 MW campus in a top-quartile European geography, with a scarcity premium precisely *because* the corridor is saturated. Scarcity value in a proven market is a real phenomenon and the bull case's genuine insight: the same saturation that blocks Atlas would, if Atlas got through, make the position unusually valuable.

Why Atlas does not underwrite it. The chain has six sequential conditions. Atlas assesses two of them — Fingrid headroom and affordable land control — as unlikely on current evidence, and the chain fails if any single link fails. Compounding even generous individual probabilities across six conditions produces a number that does not justify the option cost, and Atlas has no mechanism to influence any of the six.

43. Base Case

The base case is what the evidence actually supports today, stated without adjustment for hope.

Dimension	Base case position
Grid	Southern Finland connection restricted to at least 2027, with demonstrated willingness to extend. Node headroom at Koria and Kymi unknown and, after six committed projects, probably small. Fingrid contracts against valid permits, which Atlas cannot obtain without land.
Land	Four parcels confirmed taken, one contracted at €48m, one in preparation by others, two carrying stale reservation flags. One parcel — Redeve's retained Myllykoski holding — plausibly available at institutional pricing.
Fiscal	+2.19 c/kWh in force since 1 July 2026, roughly €19m/year gross on a 100 MW campus. Replacement relief capped at €30m nationally, conditional, terms unconfirmed.
Demand	Level 0. Real corridor demand exists and is fully allocated to competitors.
Competition	Six developments; four institutionally-backed sponsors; all three principal contractors committed through 2027.
Atlas position	No land, no grid position, no capital commitment, no operator relationship, no asset-class track record, no identified differentiation.
Base case outcome	**Atlas cannot originate a site in this corridor. The candidate fails.**

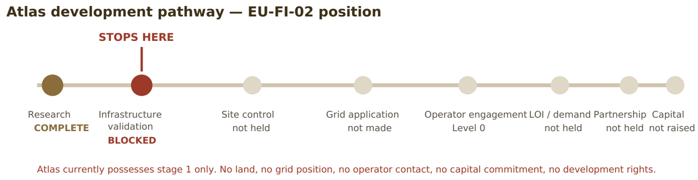
44. Bear Case

The bear case is not that the corridor fails. It is that **Atlas pursues it anyway**, and the failure mode is instructive enough to record.

- **Sunk cost accumulation.** Six to twelve months of outreach, travel and advisory spend against a market-access barrier that no amount of diligence removes. The information Atlas would buy is information the market already has.
- **Relationship burn.** Atlas has one credible first approach to each of Fingrid, Cursor Oy, Kouvola Innovation, Cinia and the major AI operators. Spending them on a candidate with no land and no grid position teaches those counterparties that Atlas is not a serious developer — a reputational cost that transfers to every future Atlas candidate in Europe.
- **The regulatory floor falls further.** Mandatory matching generation is enacted; the registration and security regime tightens; the state aid scheme excludes speculative merchant projects. Each raises the barrier in favour of incumbents.
- **The market cools.** Forecasts for Finnish capacity by 2030–31 range from 1.5 GW to over 5 GW. That spread is not precision disagreement; it is genuine uncertainty about whether the AI capex cycle sustains. A late entrant with no anchor tenant is maximally exposed to a turn.
- **The programme-level cost.** The worst outcome is not losing money on EU-FI-02. It is that Atlas concludes from a well-run FAIL that its *methodology* is wrong, when in fact the methodology worked exactly as intended and identified a correct negative in weeks rather than quarters.

45. Development Pathway

The canonical Atlas pathway, with this candidate's honest position marked. **Atlas possesses none of these stages.**



The pathway is drawn to make one point that no narrative can soften: **Atlas is at stage one of eight, and stage two is blocked by a party Atlas cannot compel and a market Atlas cannot outbid.** Nothing in this dossier should be read as implying Atlas holds land, grid capacity, permits, financing, customers or development rights in Finland. It holds none of them.

46. Final Recommendation

GATE 1 · CANDIDATE EU-FI-02 · KOUVOLA–KOTKA, FINLAND

FAIL

Close the candidate. Retain as a reference geography. Reallocate the validation budget.

Should Project Atlas invest further time and resources into actively developing Kouvola-Kotka as a potential European AI infrastructure location?

No. The corridor is a genuinely excellent location for AI infrastructure and a genuinely poor opportunity for Atlas. The distinction is not semantic: it is the difference between a market assessment and a capital allocation decision, and Atlas exists to make the second, not to admire the first.

The one thing we need to know next

THE SINGLE MOST IMPORTANT UNRESOLVED ISSUE

Does Atlas have any route to differentiated site access in *any* European geography — and if not, is site origination the right business at all?

This is deliberately not a Kouvola question, because the Kouvola questions are answered. EU-FI-02 has established something more valuable than a corridor assessment: that the Atlas just-transition site-selection thesis is **correct and non-proprietary**. Four separate parties executed it in this corridor before Atlas arrived, and a specialist landowner built a business on it. A methodology that is right and widely known generates no excess return.

That finding has three implications the Investment Committee should weigh before authorising the next candidate.

First, screening criteria must change. Atlas has been screening on *physical suitability* — power, land, cooling, fibre, heat. Every candidate that scores well on those criteria will, by construction, already have been found by better-capitalised developers, because those criteria are public and the developers are not stupid. Atlas should screen instead on **access asymmetry**: published node-level grid capacity regimes (France's pre-allocated 18 GW and published powered sites being the clearest European example), municipalities with land and no developer relationship, and geographies at an earlier point in the same conversion cycle this corridor has already completed.

Second, the honest strategic question is whether Atlas should originate at all. The corridor demonstrates a functioning value chain — landowner (Redeve), developer (atNorth, Hyperco), operator (DayOne), tenant (ByteDance), contractor (YIT, Caverion, Eltel), capital (Partners Group, Damac, Brookfield). Atlas has no capital and no track record, which are the two entry requirements for the developer role. It does have a rigorous, falsification-first research methodology that produced this dossier in weeks — and the corridor's history suggests that the party who systematically *knew first* captured real value. Whether that is a business, and whether it is a business Alejandro wants, is a Gate 0 question about Atlas rather than a Gate 1 question about Finland. It should be answered before EU-FI-03 is opened.

Third, the programme is working. This candidate was screened, researched, red-teamed and rejected without land, capital, travel or a single counterparty approach. That is the falsification-first methodology functioning exactly as designed. The failure would have been an enthusiastic dossier arguing that a 400 kV substation and a closed paper mill constitute an opportunity.

Retain, do not delete

Kouvola–Kotka should stay in the Atlas database as **reference candidate REF-01**. It is the clearest available worked example of a just-transition industrial region converting to AI infrastructure, with a documented sequence: mill closure (2006–2023) → specialist landowner acquires and holds the estate → hyperscaler anchors the region (Google, 2009) → developers arrive (2023–2024) → tenants commit (2025) → land prices reprice (€48m for 21 ha, 2026) → grid closes (2025–2027) → fiscal regime tightens (2026). **Any future Atlas candidate can be located on that timeline.** The value of EU-FI-02 to the programme is that it dates the stages.

Appendices

Appendix A — Evidence Register

Selected records. Confidence classes as defined in Document Control. Full source detail in Appendix B.

Evidence ID	Category	Claim	Confidence	Implication
FI02-PWR-002	Power	Fingrid projects Finnish consumption 86 TWh (2025) → 103–123 TWh (2030) → up to 159 TWh (2035), driven principally by data centres	CONFIRMED	National demand growth is real; so is the grid strain it causes
FI02-PWR-006	Power	Emission factor for Finnish electricity 32 gCO2/kWh (H1 2026)	CONFIRMED	ESG target achievable on annual matching
FI02-PWR-007	Power	Finnish consumption H1 2026: 46.1 TWh, +6.3% y/y	CONFIRMED	Demand outrunning reinforcement
FI02-PWR-008	Power	H1 2026 connections: 510 MW new renewable capacity (108 MW wind, 402 MW solar)	CONFIRMED	Solar now outpacing wind additions
FI02-PWR-011	Power	Finnish wholesale average fell €46/MWh (2024) → €41/MWh (2025)	STRONGLY IND.	Cost base favourable; wholesale is a minority of delivered cost
FI02-PWR-012	Power	447 negative-price hours in 2025, down 38% from 724 in 2024	STRONGLY IND.	Flexibility arbitrage narrowing
FI02-PWR-015	Power	>30,000 MW of grid energy storage connection requests; c.1,800 MW contracted	CONFIRMED	Storage competes with load for the same connection capacity
FI02-GRD-001	Grid	Koria 400/110/20 kV substation, Kouvola; 1970s build, refurbished Nov 2016–end 2019; serves Kymenlaakso and Päijät-Häme reliability	CONFIRMED	The corridor's transmission anchor; recently modernised
FI02-GRD-003	Grid	Kymi 110 kV substation north of Kotka expanded 2020 with two line bays for a customer substation	CONFIRMED	Precedent for customer-funded connection at Kotka
FI02-GRD-006	Grid	Fingrid restricted connection of new industrial consumption to the main grid in southern Finland 2025–2027	BLOCKER	**Primary blocker**
FI02-GRD-007	Grid	Fingrid unit manager: no new customers on the west coast or southern Finland connectable before 2027 at the earliest	CONFIRMED	Confirms severity
FI02-GRD-008	Grid	Fingrid has received enquiries for c.500,000 MW of production and consumption connections vs c.15,000 MW peak consumption	CONFIRMED	Queue is speculative and enormous
FI02-GRD-009	Grid	Eltel c.€16m turnkey 110 kV substation and cabling from Hyperco Kouvola site to the Fingrid substation; works from summer 2025	CONFIRMED	Real-world connection cost benchmark
FI02-GRD-011	Grid	Fingrid main grid development plan 2026–2035: c.€5.2bn; €1.7bn in 2025–2028	CONFIRMED	Reinforcement is funded but sequenced
FI02-GRD-012	Grid	Hitachi Energy to supply Fingrid seven 400 MVA / 400 kV transformers plus option for two, made in the Vaasa region	CONFIRMED	TSO absorbing domestic transformer capacity
FI02-GRD-013	Grid	No new grid storage >5 MW in southern Finland before 2029, pending Hikiä–Toivila	BLOCKER	Closes the BESS mitigation route; evidences restriction extension
FI02-GRD-019	Grid	Fingrid connection agreement requires a legally valid zoning plan and building permit	CONFIRMED	**Sequencing forecloses a landless developer**
FI02-GRD-023	Grid	KJV2026 draft published 12 June 2026; consultation to 21 Aug 2026; separate "Technical requirements for large demand facilities" applies immediately to direct connectees	CONFIRMED	New compliance regime, unquantified cost
FI02-GRD-024	Grid	≥30 MW demand facilities must withstand disturbances (fault ride-through) and be able to limit power intake	CONFIRMED	Design and capex implications
FI02-LND-001	Land	Redeve sold 21 ha at Myllykoski to atNorth (agreement 4 Dec 2023); retains >50 ha in the industrial park	STRONGLY IND.	**The only plausibly available large parcel**
FI02-LND-003	Land	Hyperco Koria: 24 ha industrial plot purchased 2024 from the City of Kouvola	CONFIRMED	Municipal land moves quickly to credible developers
FI02-LND-004	Land	Novagen Data Centre Oy preliminary agreement 20 Feb 2026 to acquire c.21 ha of the former UPM Voikkaa mill from Redeve for c.€48m; €200,000 deposit paid	STRONGLY IND.	Land repricing; last large estate contracted
FI02-LND-005	Land	Data centres under construction or in preparation in Kouvola at Myllykoski, Ummeljoki, Koria and Kuusankoski (Yle, Mar 2026)	STRONGLY IND.	Four clusters in one municipality
FI02-LND-007	Land	Kouvola Savontalo listing: 10 ha, 22,800 m²	UNVERIFIED	**Vendor claim committing three prohibited inferences**

		building rights, 30 MW short-term / >100 MW long-term, zoned social/educational, re-zoning "within a year"		
FI02-LND-009	Land	Kotka Mussalo DC Campus: 26 ha, City of Kotka, solid rock/moraine, levelled by owner, 3 km to Kotka centre, 6 km to E18; listed reserved until 04/2024	STATUS UNKNOWN	Last identified municipal coastal parcel
FI02-LND-010	Land	Google acquired 50 ha adjacent to Hamina for €4.3m (2022) for possible expansion	CONFIRMED	Incumbent land banking
FI02-LND-012	Land	Loviisa City Council approved land lease and preliminary agreement with Nova Complex 21 Jan 2026; c.6 ha phase 1; plot reservation granted autumn 2025	CONFIRMED	Template for municipal plot reservation route
FI02-LND-014	Land	Mussalo minimum recommended building elevation 314 cm; estimated minimum plot level +500 cm (FMI 2014 study basis)	INDICATED	Flood risk addressed with unusual rigour
FI02-FBR-002	Fibre	Cinia network c.17,000 km, positioned along railways, ITU-T G.652.D compliant	CONFIRMED	Kouvola rail junction implies likely presence — not proven capability
FI02-FBR-006	Fibre	C-Lion1 landing points: Helsinki and Hanko (Finland) and Rostock (Germany); c.1,173 km; 8 fibre pairs; 144 Tbit/s; c.€100m	CONFIRMED	**The corridor is not a cable landing geography**
FI02-FBR-008	Fibre	C-Lion1 fault 18 Nov 2024 in Swedish EEZ east of Öland; repaired 28 Nov 2024; characterised as sabotage by German defence minister	CONFIRMED	Baltic route risk is demonstrated, not theoretical
FI02-WTR-001	Water	Google Hamina cooled entirely with Baltic seawater since 2011, reusing paper mill intake infrastructure; pipes up to 2 m diameter	CONFIRMED	Seawater cooling proven — at a site Atlas cannot access
FI02-WTR-002	Water	Google reports PUE c.1.1 at Hamina	CONFIRMED	Free-cooling benefit quantified
FI02-HEA-001	Heat	Google–Haminan Energia heat recovery: recovered heat estimated at c.80% of Hamina's district-heating demand; contributing from late 2025; 97% carbon-free	CONFIRMED	Hamina's heat network is effectively saturated
FI02-HEA-002	Heat	atNorth FIN04 heat reuse via KSS Energia heating plant; committed publicly by KSS CEO	CONFIRMED	Kouvola's heat network is partnered
FI02-MKT-001	Market	Google bought the Summa mill and most of the site from Stora Enso for c.€40m (agreed 2008, closing Q1 2009); mill closed Jan 2008 after 53 years	CONFIRMED	The corridor's founding conversion precedent
FI02-MKT-004	Market	atNorth FIN04: 430 MW target across 45 ha; YIT design-build c.€300m agreed 21 July 2026, completion end-2027	CONFIRMED	Scale and momentum of the leading competitor
FI02-MKT-006	Market	Hyperco Koria: 24 ha, c.28,000 m², TikTok/ByteDance tenant, YIT main contractor from end-Oct 2025, completion May 2027; Caverion MEP agreement 10 Dec 2025 among the largest in Assemblin Caverion Group history	CONFIRMED	Anchor tenancy already allocated
FI02-MKT-007	Market	Hyperco filed June 2026 for a second Koria data centre: 100 MW across 56,670 m², with an early-start request	STRONGLY IND.	Further headroom consumption
FI02-MKT-009	Market	DayOne Lahti: c.€1.2bn, 98,901 m² fully zoned industrial, 128 MW IT total with a 50 MW first building; construction from Q3 2025; RFS 2027; c.1,000 construction workers at peak; tenant negotiations unconcluded as at Aug 2025	CONFIRMED	Adjacent competition; labour draw
FI02-MKT-013	Market	Stora Enso decided June 2023 to close the Sunila pulp mill (Kotka) and one of two paper machines at Anjala; c.1,150 group-wide reductions, c.600 at affected facilities	CONFIRMED	Source of the brownfield estate
FI02-MKT-014	Market	Finnish Government granted Kymenlaakso structural change support (€1m to employment and business services), citing closures plus Russian border effects	CONFIRMED	Political receptiveness — and economic fragility
FI02-MKT-019	Market	Confederation of Finnish Industries / FDCA 2025 study: Finnish DC market c.285 MW, potentially >1.5 GW by 2030; announced projects >3.4 GW	STRONGLY IND.	Best-sourced market sizing; adopted
FI02-REG-001	Regulation	Data centre electricity tax moved from category II (0.05 c/kWh) to category I (2.24 c/kWh) from 1 July 2026; +2.19 c/kWh; c.€47m annual revenue at 2026 levels	CONFIRMED	**Second blocker**
FI02-REG-002	Regulation	Category I total 2.917875 c/kWh and category II 0.169425 c/kWh incl. 0.085 c/kWh stockpile fee and 25.5% VAT; stockpile fee raised 1 Apr 2026	CONFIRMED	Precise delivered tax lines
FI02-REG-003	Regulation	Fixed-term ten-year state aid tax relief based on electricity consumed, max €30m revenue loss; legislation in force 1 July 2026; conditional on	CONFIRMED	Partial mitigation, conditional and capped

		registration plus waste heat reuse or above-baseline efficiency; EU state aid compatibility to be taken into account; security authorities to review the sector		
FI02-REG-006	Regulation	Google linked future Finnish expansion decisions to the tax settlement; XTX Markets reported reviewing a >$1.1bn Kajaani investment	INDICATED	Incumbents themselves repricing Finland
FI02-REG-007	Regulation	Kokoomus legislative proposal, reported SDP-supported, would require large data centres to secure matching new generation before connecting	INDICATED	Forward regulatory risk; status unverified
FI02-REG-008	Regulation	Where zoning does not permit data centre use an amendment is required — "often lengthy, potentially taking years and involving multiple rounds of appeals"; projects typically sited near substations on land already permitting the use	CONFIRMED	Refutes the Savontalo one-year re-zoning claim
FI02-REG-010	Regulation	Non-EU/EEA purchasers need a Ministry of Defence permit where ownership or effective influence exceeds 10%	CONFIRMED	Capital structure constraint
FI02-REG-011	Regulation	France pre-allocated 18 GW of grid capacity for data centres with first-ready-first-served rolling out in 2026, plus a fast-track and named powered sites for gigawatt-scale projects	INDICATED	**Benchmark for an accessible allocation regime**
FI02-LOG-001	Logistics	HaminaKotka: Finland's largest universal and container port, formed 2011, c.1,100–1,200 ha, c.1m TEU Mussalo capacity, direct E18 and rail, specialises in demanding project shipments	CONFIRMED	Genuine corridor advantage for heavy equipment
FI02-WRK-001	Workforce	Kymenlaakso population 186,030 (1990) → 175,377 (2010) → 162,812 (2020) → 156,198 (2025)	CONFIRMED	Structural labour constraint
FI02-WRK-007	Workforce	atNorth reported c.100 applications after one Kouvola recruitment event; c.60 staff at FIN04 phase 1, c.200 later	CONFIRMED	Recruitment feasible at current scale
FI02-WRK-008	Workforce	Xamk: Finland's third largest UAS, c.12,000–12,300 students, >1,000 staff, campuses at Mikkeli, Kotka, Kouvola, Savonlinna; offers MEng in Digital Industry and AI	CONFIRMED	Credible technician and engineer pipeline
FI02-WRK-010	Workforce	Deloitte impact study for Google Hamina expansion: c.1,900 direct, c.1,650 indirect, c.640 induced construction jobs annually over two years	CONFIRMED	Scale of the concurrent labour draw
FI02-DEM-001	Demand	TikTok €1bn Kouvola investment via HyperCo Oy as part of the €12bn Project Clover	CONFIRMED	The corridor's anchor demand — fully allocated
FI02-DEM-002	Demand	Fingrid's pause on large connections reportedly affected data centre projects at Akaa Point requiring 10 MW or more, lasting to 2027	INDICATED	A comparable small AI operator hit the same constraint
FI02-FIN-001	Finance	Brookfield, alongside a global sovereign wealth investor, provided a c.€1bn holdco financing for DayOne Data Centers, advised on Finnish law aspects	CONFIRMED	The calibre of capital Atlas would compete against

Appendix B — Source Register

Sources consulted and cited. Tier 1: TSO, government, regulator, municipality, company primary announcement. Tier 2: established business and trade press. Tier 3: consultancy, marketing, aggregator, commercial database.

#	Tier	Source	Subject
1	1	Fingrid — "The connection of electricity consumption facilities to the main grid is temporarily tight in southern Finland"	2025–2027 connection restriction
2	1	Fingrid — "Fingrid secures transmission capacity for growth… restrictions on new energy storage facility connections continued in southern Finland"	Restriction extension to 2029; 30 GW storage requests
3	1	Fingrid — "Fingrid improves connection planning and information flow"	Approval procedure; 2–10 MW thresholds
4	1	Fingrid-lehti — "New provisions for grid connections"	Parviainen statement; 500,000 MW enquiries; permit precondition
5	1	Fingrid — Korian sähköaseman uudistaminen (project page)	Koria 400/110/20 kV substation
6	1	Fingrid — Kymin sähköasema (project page)	Kymi 110 kV expansion 2020
7	1	Fingrid — Hankkeet / project archive index	Kymin johtojärjestelyt; Kyminlinna–Pernoonkoski
8	1	Fingrid — "Main grid development plan: investments in electricity transmission links continue to grow"	2026–2035 plan; eastbound 400 kV EIA
9	1	Fingrid — "Main grid development plan updated: reinforcing the grid creates growth opportunities"	Regional plans added
10	1	Fingrid — Construction / grid development pages	€5.2bn investment programme
11	1	Fingrid — Group half-year report 1.1.–30.6.2026	Consumption 46.1 TWh; 32 gCO2/kWh; Grid Scope; DLR
12	1	Fingrid — "Draft for comments: KJV2026 Grid code specifications for demand connections"	KJV2026; large demand facility requirements
13	1	Fingrid — Grid code specifications for demand connections	Technical requirements document scope
14	1	Fingrid — "Temporary restrictions on grid connections for grid energy storage… west coast"	Single bidding zone; reinforcement dates
15	1	Fingrid — main grid development plan 2022–2031 (PDF)	400 kV reinforcement needs; DLR
16	1	Finnish Government / Prime Minister's Office — "Government safeguards Finland's competitiveness in attracting data centre investments"	Tax move to category I from 1 July 2026; €47m
17	1	Finnish Government — "Ministerial Committee on Economic Policy decides on state aid for data centres and registration obligation"	10-year relief; €30m cap; conditions; security review
18	1	Finnish Government — "Government grants support to Kymenlaakso to manage structural change"	Sunila and Anjala closures; €1m support
19	1	City of Kotka — structural change support page	Regional restructuring
20	1	Ministry of Finance / Eduskunta — Bill HE 156/2025 vp	Repeal of electricity tax provisions
21	1	Business Finland — "Google announced an expansion of its Hamina data center"	€1bn; €3.5bn cumulative; Deloitte job study
22	1	Business Finland — Explore business opportunities: data centers site map	Kotka Mussalo; Kouvola Savontalo; Kuusaa; Ahvenkoski
23	1	Google — Hamina data centre location page	Seventh facility; €3.5bn; 98% CFE 2023; heat recovery
24	1	YIT Corporation — investor news, 21 July 2026	atNorth Myllykoski c.€300m design-build; 430 MW campus
25	1	YIT — Kouvola data centre project page	Hyperco Fin HoldCo 1 partnership
26	1	Caverion — "Kouvola data center built with Finnish expertise…"	10 Dec 2025 agreement; MEP scope; Taipale and Tamminen quotes
27	1	Eltel — "Eltel signs Data Center contract with Hyperco valued at EUR 16 million"	110 kV substation and cabling to Fingrid substation
28	1	atNorth — "atNorth announces heat reuse enabled mega site in Kouvola"	FIN04; 21 ha; 60 MW phase 1; KSS Energia; Riipinen and Toikka quotes
29	1	atNorth — Myllykoski data center page	Free air cooling; DLC; KSS heat recycling
30	1	Redeve — "Billion-dollar atNorth data centre to be built in the former Myllykoski mill area owned by Redeve"	4 Dec 2023 agreement; 21 ha sold; >50 ha retained; Lahtinen

31	1	TikTok Newsroom — "Project Clover: €1 Billion Investment in Finland for New Data Center"	HyperCo Oy; €12bn Project Clover; Toikka quote
32	1	DayOne — "DayOne announces flagship hyperscale data center project in Lahti, Finland"	€1.2bn; 128 MW; 98,901 m²; 50 MW first building; Khoo quote
33	1	DayOne — Finland market page	281 MW; Lahti and Kouvola; zero freshwater; Hyperco partnership
34	1	Port of HaminaKotka — corporate pages (containers; multipurpose port; about)	Largest Finnish port; Mussalo; project shipments; E18 and rail
35	1	Cinia — International connectivity / fibre optic network	17,000 km; railway alignment; G.652.D; C-Lion1
36	1	Telia Finland — Data Center Services connectivity; Telia Wholesale network	Four fibre routes; carrier list; 99% coverage
37	1	Xamk — Kotka campus / study pages	Regional context; campus scale
38	1	ymparisto.fi — Fingrid 400 kV EIA procedures (Hausjärvi–Anttila; Anttila–Länsisalmi)	YVA process structure; ELY Centre role
39	1	Powercoast.fi / Business Kotka-Hamina — Nova Complex Loviisa articles	Land lease 21 Jan 2026; 6 ha; plot reservation; Lindström and Mah quotes
40	1	Business Kotka-Hamina — data centre business and plots; region pages	Three pre-zoned areas; regional positioning
41	1	Statistics Finland — municipality and sub-region classifications 2026; population publications	Kouvola 77,625; Kotka 50,029; Hamina 19,113; Kymenlaakso 156,198
42	2	Datacenter Dynamics — "AtNorth files to expand planned data center campus in Kouvola"	430 MW / 45 ha; Ummeljoki permit; Yle-sourced timing
43	2	Datacenter Dynamics — "Hyperco files to build another data center in Kouvola"	100 MW second building; 56,670 m²; 16 June 2026 committee
44	2	Datacenter Dynamics — "Former paper mill site in Finland targeted for data center use"	Novagen; Voikkaa; c.20 ha; €48m
45	2	Datacenter Dynamics — "Damac's Hyperco targets data center development in Espoo"	Edgnex acquisition; Lohja 300 MW; Pyhäjoki 130 ha
46	2	Datacenter Dynamics — "Google launches heat recovery project at data center in Hamina"	€1bn; 50 acres 2022; capacity undisclosed
47	2	Datacenter Dynamics — "Plans on hold for Google data center in Finland following potential electricity tax rise"	Google position; XTX review; FDCA 285 MW → 1.5 GW
48	2	Datacenter Dynamics — "AI cloud company DataCrunch proposes data center in Akaa"	Fingrid pause affecting ≥10 MW projects to 2027
49	2	Datacenter Dynamics — "atNorth announces 60MW+ data center in Finland — with heat reuse"	Kristinsson quote on AI operator discussions
50	2	Datacenter Dynamics — "Cinia opens submarine cable between Germany and Finland"	C-Lion1 specification and positioning
51	2	Yle — "Kouvolan ensimmäiset datakeskukset otetaan käyttöön jo muutaman kuukauden kuluttua"	Myllykoski, Ummeljoki, Koria, Kuusankoski; Ahti-Aalto and Vaarala
52	2	Yle — "Stora Enso announces plans for plant closures, 900 more job cuts"	Sunila closure; 1,150 reductions
53	2	Yle — "Stora Enso closes Kotka mill, cuts 240 jobs"	Kotka mill
54	2	Yle News — "Telecoms cable break reported between Finland and Germany"	C-Lion1 fault; Traficom comment; 144 Tbit/s
55	2	Montel — "Finnish cut to data centre subsidy risks investment – industry"	Purra announcement; Turtiainen and Ruusunen positions; €30m
56	2	Montel — "Finland to replace data centre subsidy with new scheme next year"	1 July 2026 date
57	2	Computer Weekly — "Nordic datacentre investments surge amid tightening licence rules"	Kokoomus matching-generation proposal; Perholehto (SDP)
58	2	Computer Weekly — "Finland government tax proposal worries datacentre investors"	€50m 2025 tax break value; MoF process
59	2	Bloomberg Tax — "Finland Parliament Considers Bill to Increase Electricity Tax Rate for Data Centers"	HE 156/2025 vp accepted for consideration 23 Oct 2025
60	2	Reuters via Lightwave / Euronews — Google Hamina investment coverage	Historic investment tranches
61	2	Pulp & Paper News — "€48M deal may turn former Finnish paper mill into huge data hub"	Voikkaa; Redeve; Tyagi; 20 Feb 2026; €200,000 deposit; c.30 companies on site
62	2	Pulp & Paper News — "UPM Explores Data Centre Future for Former Paper Mill in Finland"	Kaukas, Lappeenranta — adjacent conversion activity
63	2	Datacenter Forum — Google Hamina expansion and land acquisition articles	50 ha for €4.3m; Ikonen quote; Verne Mäntsälä; AmpTank Utajärvi

64	2	Data Centre Magazine — "Google Expands Data Centre Footprint with Finnish Land Deal"	1,400 ha from Metsähallitus; Orpo quote
65	2	Data Centre Magazine — atNorth Kouvola announcement coverage	FIN04 positioning
66	2	BeBeez International — "Hiljaa hyvä tulee: Finland's data center boom"	FDCA 285 MW / 1.5 GW / 3.4 GW; tax mechanics
67	2	Castrén & Snellman — "Fingrid extends restrictions on new energy storage facility connections in Southern Finland until 2029"	>5 MW storage restriction; Hikiä–Toivila; flexible connections; DayOne €1bn financing
68	2	DLA Piper / Lexology — "Status of data centre projects in Finland"	HE 156/2025 vp; zoning amendment duration; MoD permit; Mattila roadmap
69	2	Mondaq — Finnish storage connection restriction commentary	Corroboration of source 67
70	2	ArcticToday / YIT release syndication	atNorth Myllykoski contract, July 2026
71	2	Transformer Magazine — "Hitachi Energy to supply Fingrid transformers"	7 × 400 MVA / 400 kV plus 2 options; €5.2bn plan
72	2	Hitachi Energy — press release, June 2026	Vaasa manufacturing; Fingrid agreement
73	2	TechCrunch — "Google Buys … A Paper Mill?"	Stora Enso Summa sale, c.€40m; mill closure rationale
74	2	Breakbulk News — "HaminaKotka Port Sees Significant Growth in Q1 2025"	3,658,725 tonnes; cargo mix
75	2	IET Generation, Transmission & Distribution (Haakana et al., 2025)	Finnish spot price series 2014–2024; €1,896/MWh peak; −€500/MWh
76	2	Telecompaper — Lounea / Elisa fibre coverage	Eastern Finland fibre appointments
77	2	Submarine Networks — C-Lion1 system pages and restoration report	Specification; repair timeline; ownership
78	2	Newsweek — C-Lion1 cable cut coverage	Landing points; Öland location; joint Finland–Germany statement
79	3	Wikipedia — C-Lion1; Kouvola; Hamina; Kymenlaakso; Kotka-Hamina sub-region; Xamk; Port of Hamina-Kotka; Sunila; Kouvola railway station	Population series; landing points; institutional facts. **Used only where corroborated or non-contentious.**
80	3	datacenterfinland.fi	Savontalo listing — treated as an unverified vendor claim
81	3	Baxtel; DataCenterMap; datacenters.com	Facility aggregation. Treated as Tier 3; not used for critical conclusions
82	3	Blackridge Research — "Top 7 Upcoming Data Centers in Finland 2026"; "Finland Removes Data Center Power Tax Breaks"	Pipeline aggregation; tax impact modelling. Contradiction C2
83	3	Research and Markets / Arizton / GlobeNewswire — Finland data centre market reports	Market sizing. Contradiction C2
84	3	Energy Tech — "Finding True North: Finland Positions Power Infrastructure to Attract Data Centers" (quoting DC Byte)	Contradiction C1 — Fingrid "accommodating" characterisation
85	3	PGE.fi — "Fingrid KJV2026: New Grid Rules for Data Centers"; technical white paper	KJV2026 interpretation; consultation dates; ≥30 MW scope
86	3	Adven — industrial electricity connection commentary	Corroboration of the connection restriction; 40 → 101 TWh forecast
87	3	Gridio — "2025 Electricity Prices: 5 Insights for Finnish EV Drivers"	2025 average €41/MWh; 447 negative hours
88	3	GlobalPetrolPrices.com — Finland electricity prices	Business all-in retail c.€0.091/kWh, Dec 2025
89	3	euenergy.live; kwhprice.eu	Generation mix; Olkiluoto-3; single bidding zone
90	3	Vaasan Sähkö — electricity tax categories page	Precise category I/II totals including stockpile fee and VAT
91	3	Substack (Czyzak) — "Data Center Grid Queues"	France 18 GW pre-allocation; Fingrid permit precondition; high-value-adding DC concept
92	3	Medium (secondary aggregation)	24,000 MW connection applications; €2.3m 400 kV connection fee; 7–8 year line timeline. **Flagged as uncorroborated**
93	3	Unisco; Wood from Finland Conference; Rauanheimo	HaminaKotka throughput and terminal detail

Appendix C — Grid Infrastructure Register

See Table 3 (§6) for the primary register. Supplementary notes below.

Item	Note
Koria — capacity	**No rating published.** Atlas does not estimate. The substation's 400/110/20 kV configuration and the recent full-duplex 400 kV switchgear renewal indicate a modern, capable asset — which says nothing about uncommitted headroom.
Kymi — capacity	No rating published. The 2020 expansion added two 110 kV line bays for one customer.
Corridor 220 kV	No 220 kV assets identified in the corridor. Finland's 220 kV network is limited relative to 400 kV and 110 kV.
Nearby generation	Loviisa nuclear c.60 km west of Kotka. Historic Kymijoki run-of-river hydro. No large corridor thermal generation identified post-industrial closures.
Reinforcement relevant to the corridor	Hikiä–Toivila (2029). Eastbound 400 kV link (EIA being launched). Neither is confirmed to relieve Kymenlaakso specifically.
Dynamic Line Rating	Fingrid states it is among the first TSOs worldwide to deploy DLR on all its 400 kV lines, using a computational solution based on actual weather conditions. Marginally increases usable transmission capacity without new build.

Appendix D — Candidate Land Register

See Table 9 (§13). Summary position: eleven areas identified; one plausibly available (Redeve retained estate, Myllykoski, >50 ha); two of unknown current status (Kotka Mussalo 26 ha, Ahvenkoski); two unverified vendor listings (Savontalo, Kuusaa); six confirmed taken, contracted or in preparation by others. **Atlas holds no option, no reservation and no agreement over any parcel.**

Appendix E — Finnish Data Centre Project Database

See Table 14 (§21). Corridor subset: Google Hamina (operating, 7 facilities, >€4.5bn); atNorth FIN04 Myllykoski/Ummeljoki (430 MW target, 45 ha, under construction); Hyperco/DayOne Koria building 1 (€1bn, TikTok, completion May 2027); Hyperco Koria building 2 (100 MW, permitting); Novagen Voikkaa (21 ha, €48m, preliminary agreement); Kuusankoski (in preparation). Adjacent: DayOne Lahti (€1.2bn, 128 MW); Nova Complex Loviisa (6 ha phase 1); Nebius/Polarnode Lappeenranta; UPM Kaukas conversion under exploration.

Appendix F — Stakeholder Database

See Table (§35). Nineteen organisational categories across transmission, regulation, distribution, municipal, regional, national promotion, regional development, energy utilities, landowners, port, fibre, construction, operators, investors, government and education.

Appendix G — Contact & Outreach Database

See Table 21 (§36). Fifteen named individuals on the public record, in professional capacity, with organisational contact routes only. **Zero email addresses are asserted in this dossier.** Atlas has deliberately declined to generate inferred contact details; see the method note at §36. Outreach sequence at §37; execution is not authorised for this candidate.

Appendix H — Risk Register

See Table 22 (§38) — 24 material risks with probability, impact, evidence, mitigation, owner and gate effect; heatmap at Figure 6. Six gate-blocking: R01 grid capacity, R02 land, R03 electricity tax, R04 demand, R05 competitive saturation, R06 state aid terms. R24 (methodology non-proprietary) is a programme-level finding.

Appendix I — Power Scenario Assumptions

Assumption	Basis and caveat
Land per MW	Derived from atNorth's stated 430 MW across 45 ha (c.9.6 MW/ha) and applied linearly across scenarios. **Linearity is an Atlas assumption**; real campuses do not scale linearly and density varies with cooling architecture and building height.
Connection voltage thresholds	Atlas engineering judgement: 110 kV to c.100 MW, 400 kV above c.250 MW. **Not a Fingrid statement.** The Hyperco Koria 110 kV precedent supports the lower band.
Transformer sizing	Inferred from typical utility practice and from Fingrid's own 400 MVA procurement. Not quoted by any supplier.
Build duration	Anchored to two observed corridor projects: Hyperco Koria (Oct 2025 → May 2027) and atNorth Myllykoski (July 2026 → end-2027). Both are single buildings, not full campuses.
Reinforcement duration	7–8 years for a new 400 kV line, from a single secondary source. INDICATED — not corroborated against a Fingrid primary statement. Treat as order-of-magnitude only.
Annual consumption at 100 MW	c.850 GWh assumed for the tax calculation, implying a high load factor. **Actual consumption depends on load factor and PUE and is not established.** The resulting c.€19m/year tax figure scales directly with this assumption.
Delivered power cost stack	Illustrative. Three components (transmission tariff, balancing, GOs) are UNKNOWN, so the c.€64/MWh subtotal is a **floor, not an estimate**.
KJV2026 compliance cost	**Not modelled.** No basis exists for an estimate.

Appendix J — Regulatory & Permitting Sources

Instrument	Status and source
Act on Excise Duty on Electricity and Certain Fuels (as amended)	Data centres in category I from 1 July 2026 via HE 156/2025 vp. Sources 16, 20, 59, 90.
Data centre state aid scheme	Ministerial Committee on Economic Policy decision, 5 December 2025; legislation in force 1 July 2026; detailed terms and EU state aid position not confirmed by Atlas. Source 17.
Data centre registration obligation	Cross-administrative legislative project launched December 2025. Source 17.
National Roadmap for Data Centres	Rapporteur Veli-Matti Mattila, appointed by PM Orpo June 2025. Sources 17, 68.
KJV2026 Grid Code Specifications for Demand Connections	Draft published 12 June 2026; consultation to 21 August 2026; Energy Authority confirmation targeted early-to-mid 2027. Sources 12, 85.
Technical requirements for large demand facilities	Published 12 June 2026; applies immediately to ≥30 MW direct Fingrid connectees. Sources 12, 13, 85.
EU Commission Regulation 2016/1388 (Demand Connection Network Code)	Basis for Fingrid's specifications with national additions. Source 13.
YVA (EIA) procedure	ELY Centre as coordinating authority; reasoned conclusion issued. Source 38.
Real estate acquisition by non-EU/EEA parties	Ministry of Defence permit where ownership or effective influence exceeds 10%. Source 68.
Matching generation requirement (proposed)	Kokoomus legislative proposal; SDP-supported; status unverified. Source 57.

REQUIRED DISCLAIMER. Project Atlas is an independent research and validation programme by Albedo Industries. Candidate inclusion does not indicate that land, grid capacity, permits, financing, customers or development rights have been secured. All infrastructure opportunities remain subject to technical, commercial, regulatory and legal due diligence.

Limitations of this dossier. Desk research only. No site visit was undertaken. No counterparty was contacted. No confidential information was used. Fingrid was not approached and no node-level capacity figure was obtained or is asserted. No environmental site assessment, geotechnical report, carrier route survey, water permit consultation or legal title review was commissioned. Several figures rest on single secondary sources and are flagged accordingly. Four material contradictions between sources are disclosed at §7 (C1), §21 (C2), §27 (C3) and §31 (C4). Readers must not treat any INDICATED, UNVERIFIED or UNKNOWN classification as a basis for capital commitment.

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